

BFS Traversal

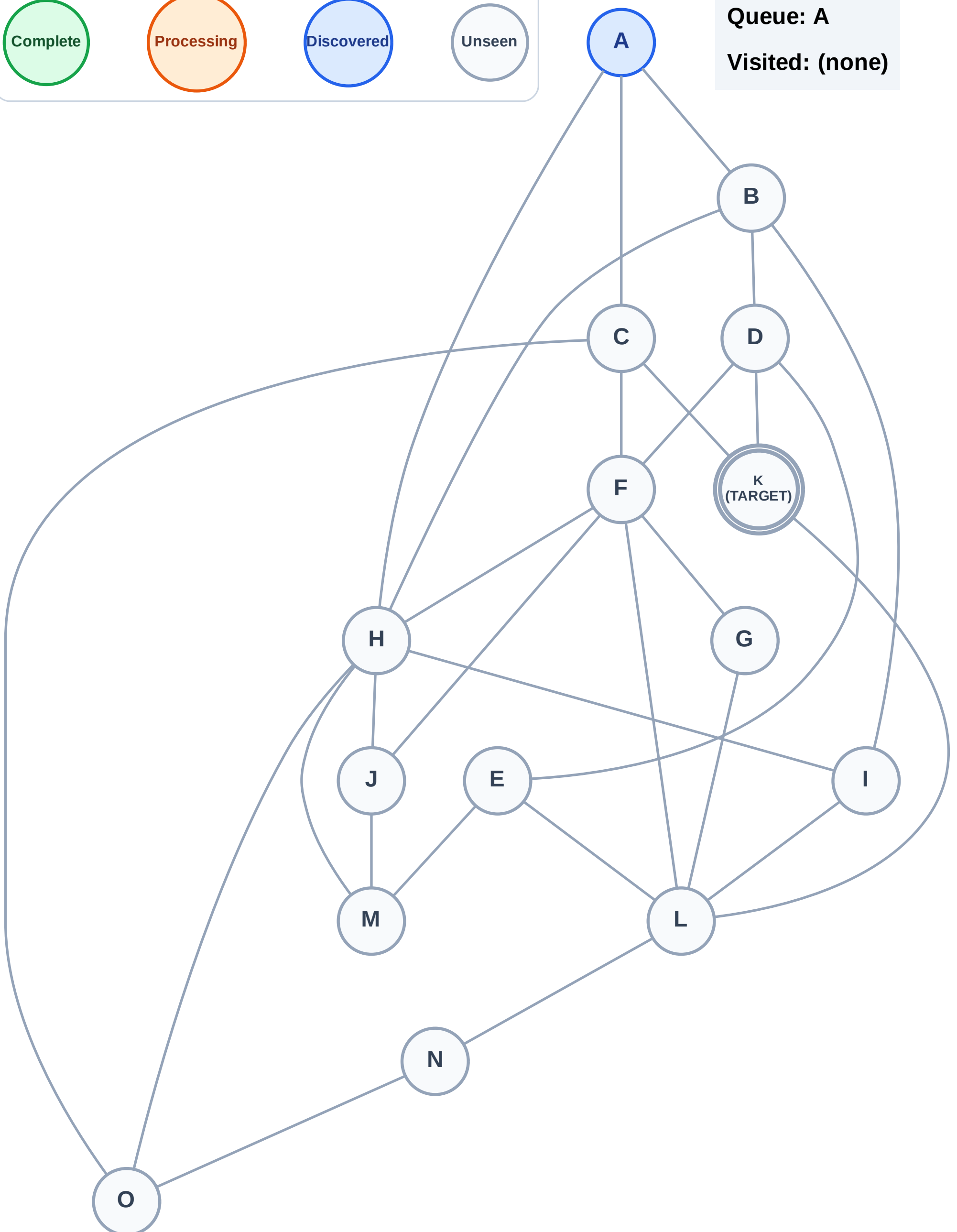
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

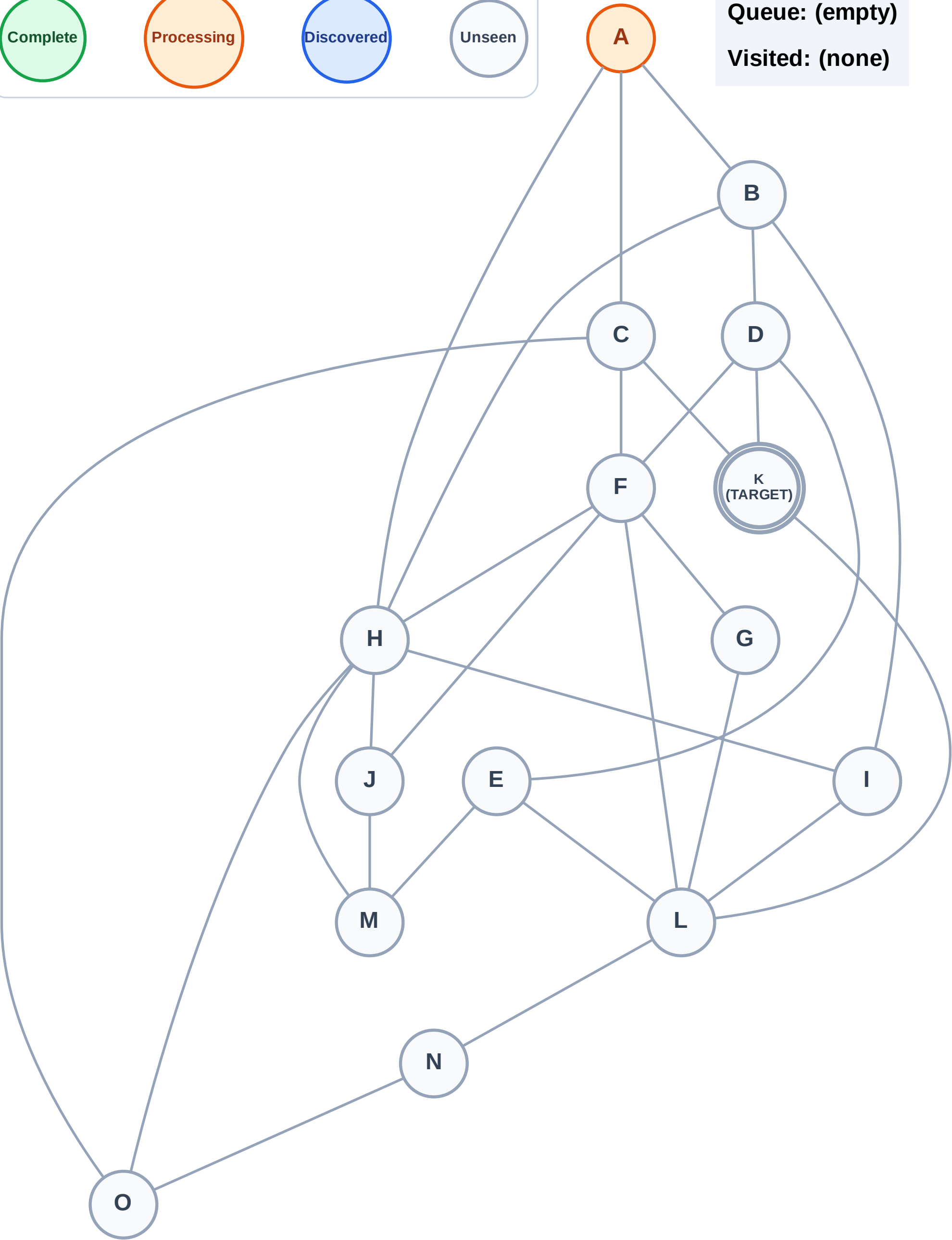
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, H from A

Legend



Complete



Processing



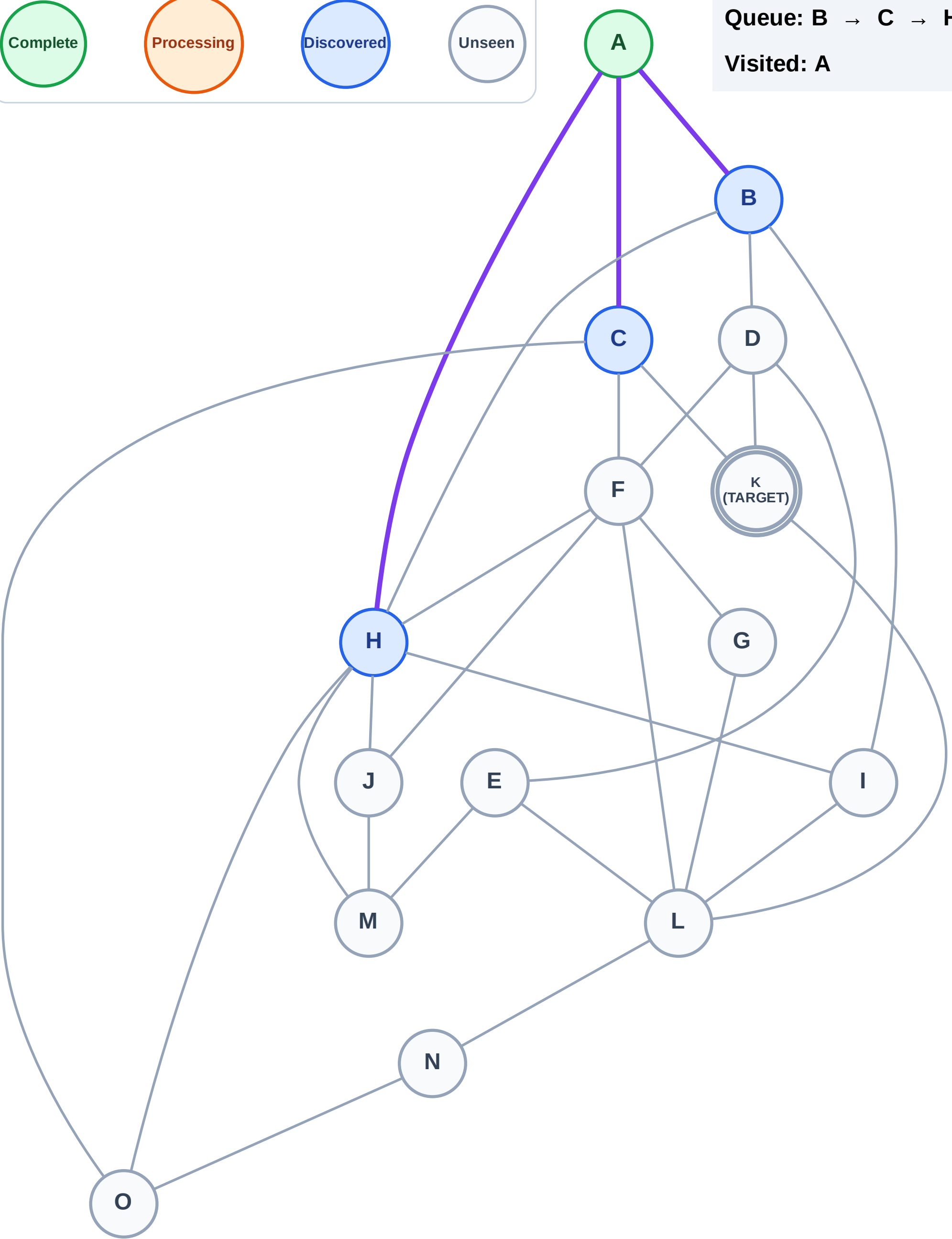
Discovered



Unseen

Queue: B → C → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

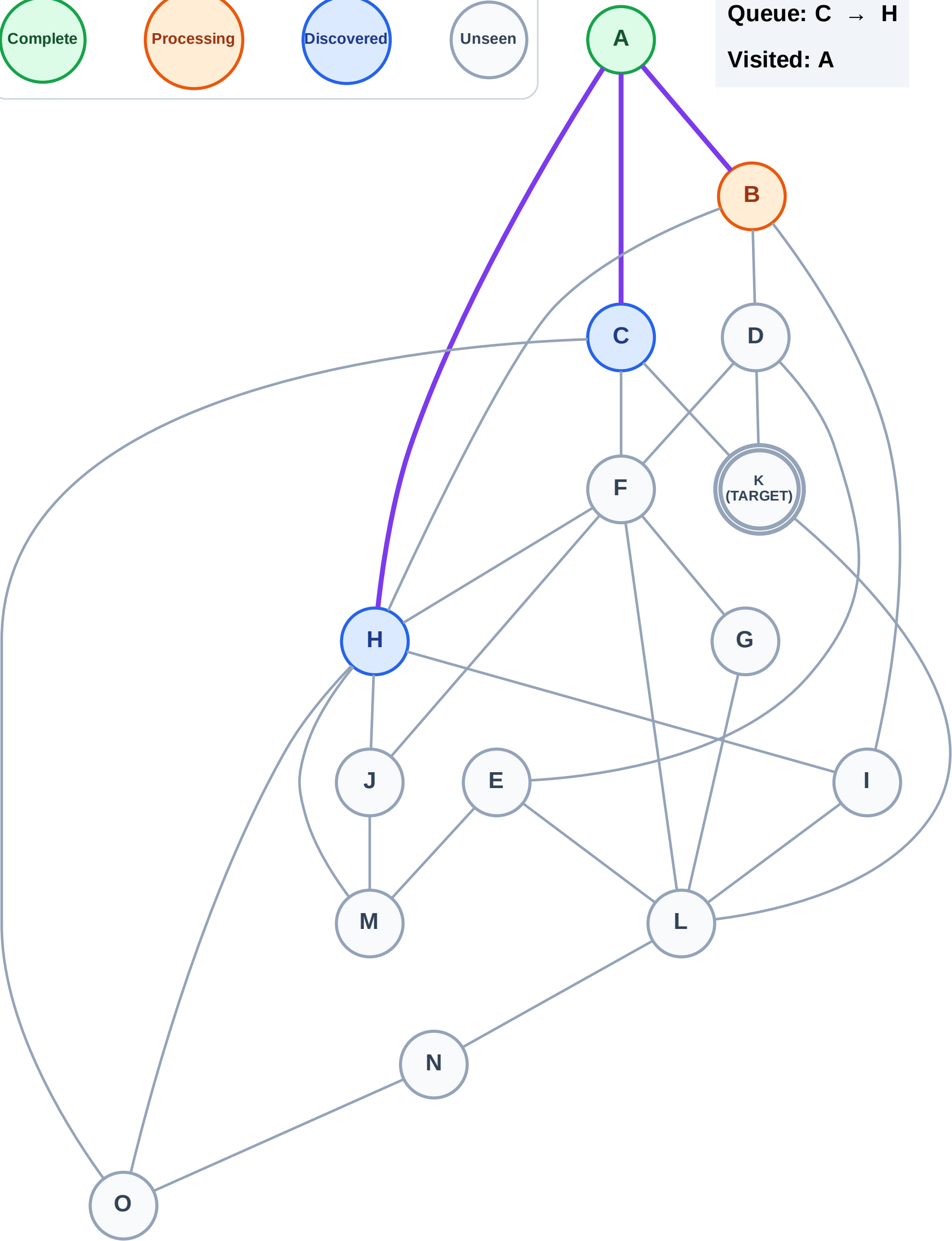
Complete

Processing

Discovered

Unseen

Queue: C → H
Visited: A



BFS Traversal

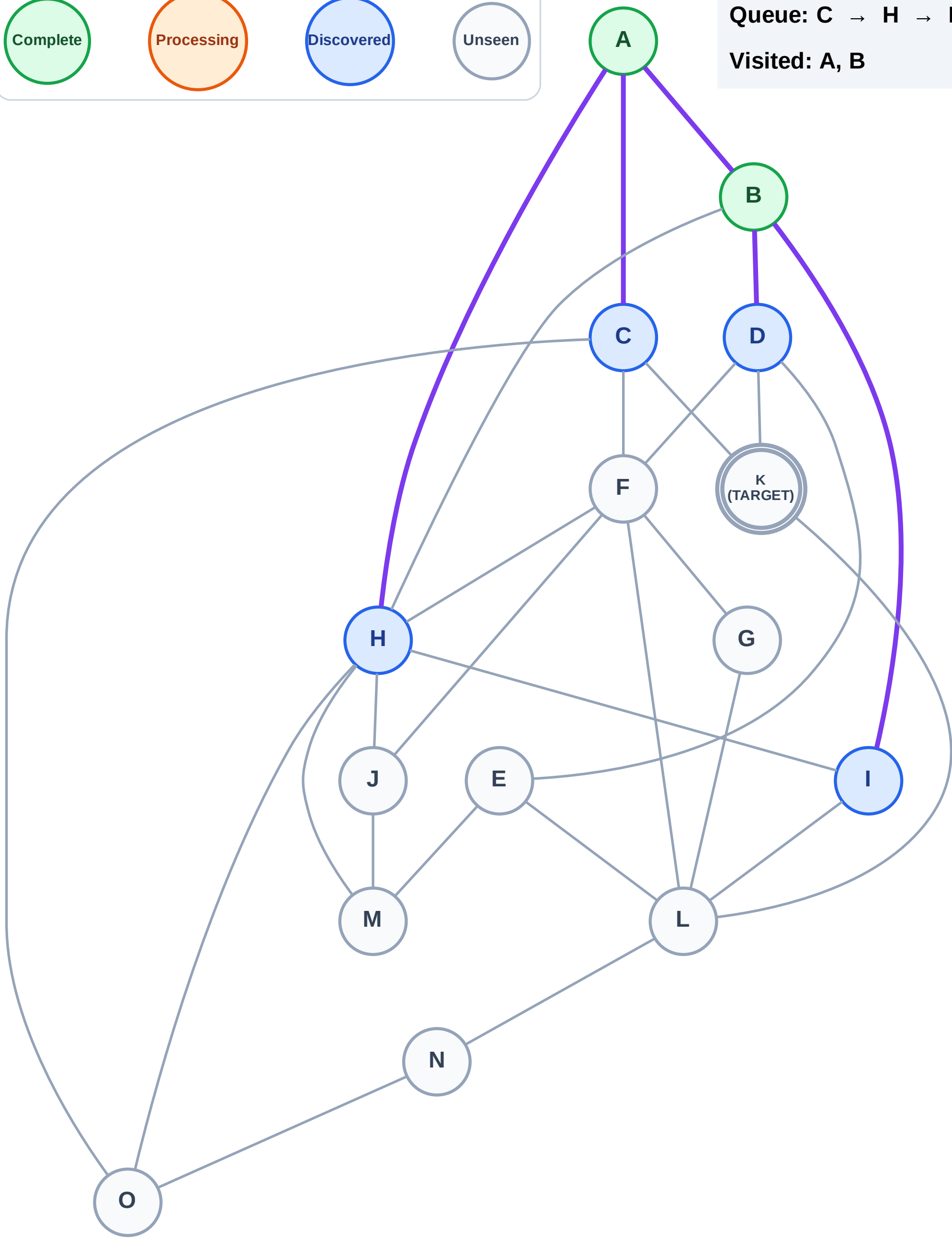
Step 5: Discover D, I from B

Legend



Queue: C → H → D → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

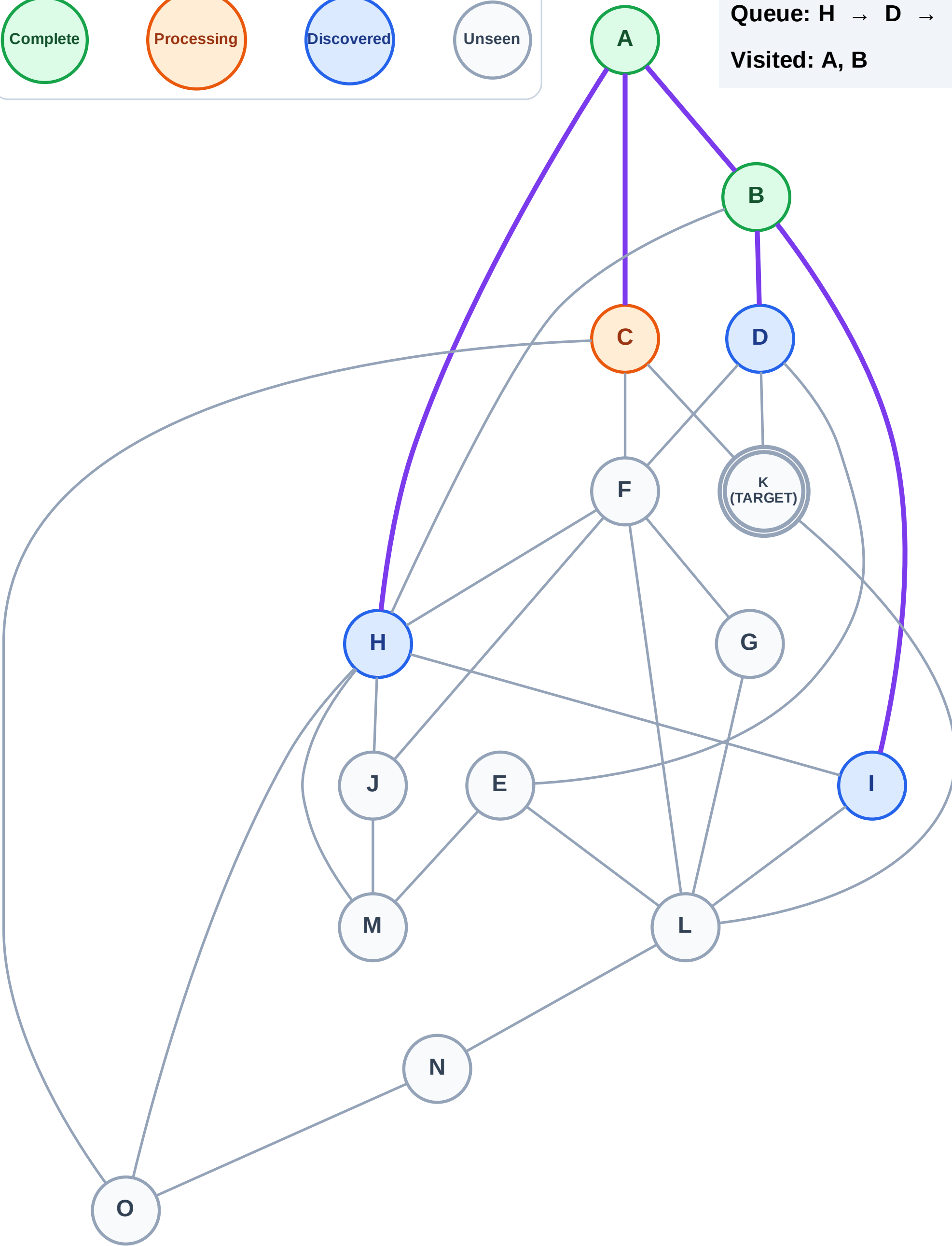
Processing

Discovered

Unseen

Queue: H → D → I

Visited: A, B



BFS Traversal

Step 7: Discover F, K, O from C

Legend

Complete

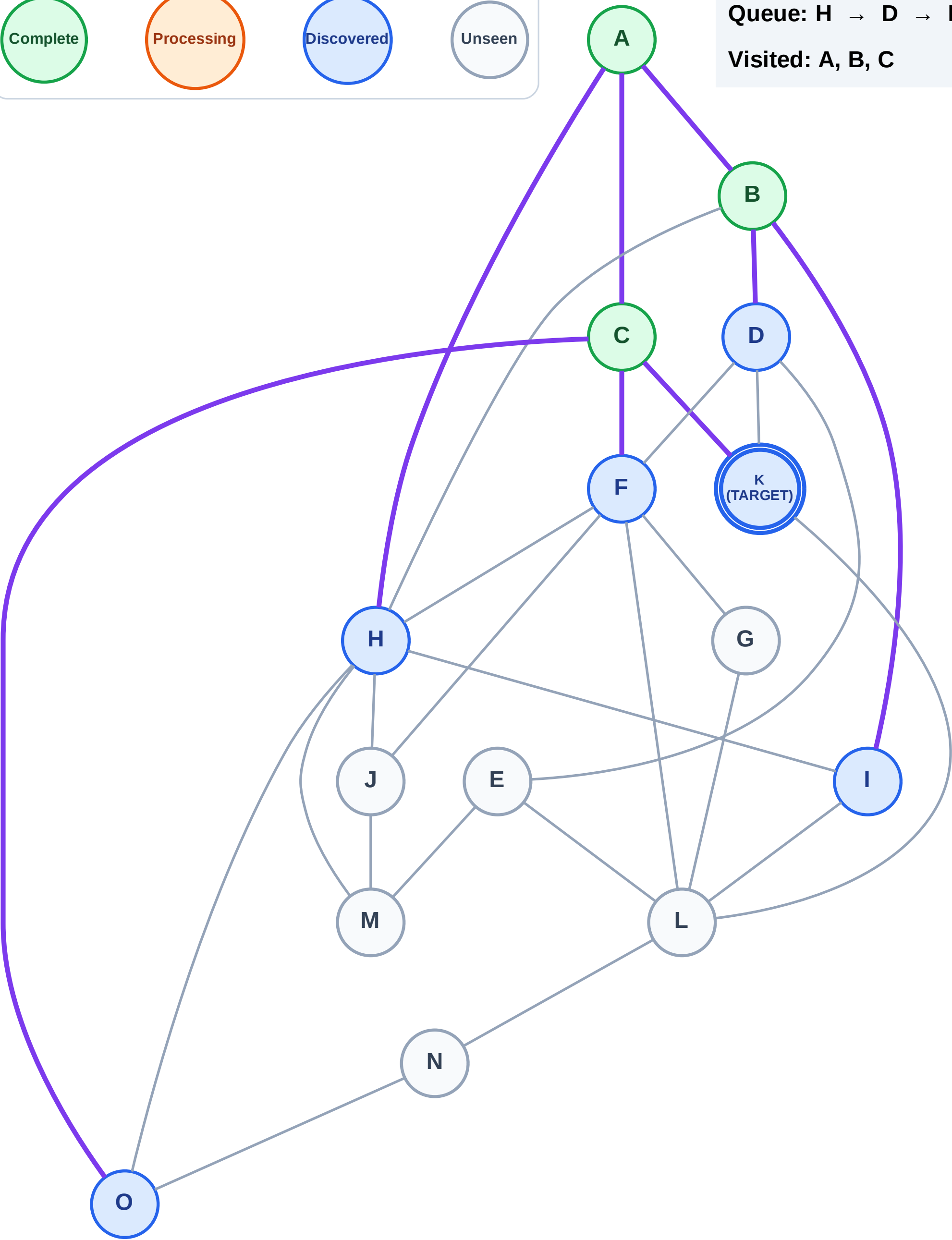
Processing

Discovered

Unseen

Queue: H → D → I → F → K → O

Visited: A, B, C



BFS Traversal

Step 8: Process node H

Legend

Complete

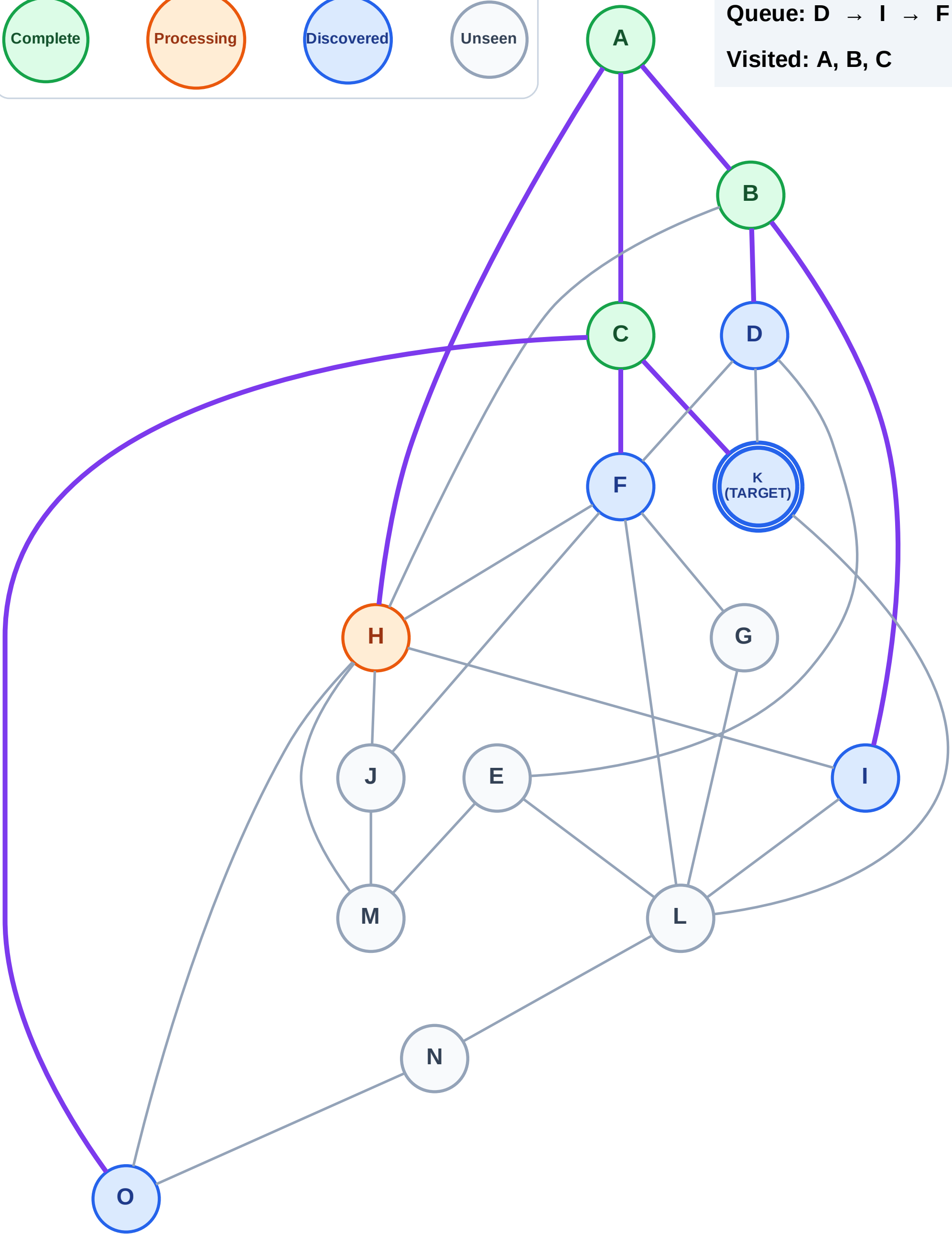
Processing

Discovered

Unseen

Queue: D → I → F → K → O

Visited: A, B, C



BFS Traversal

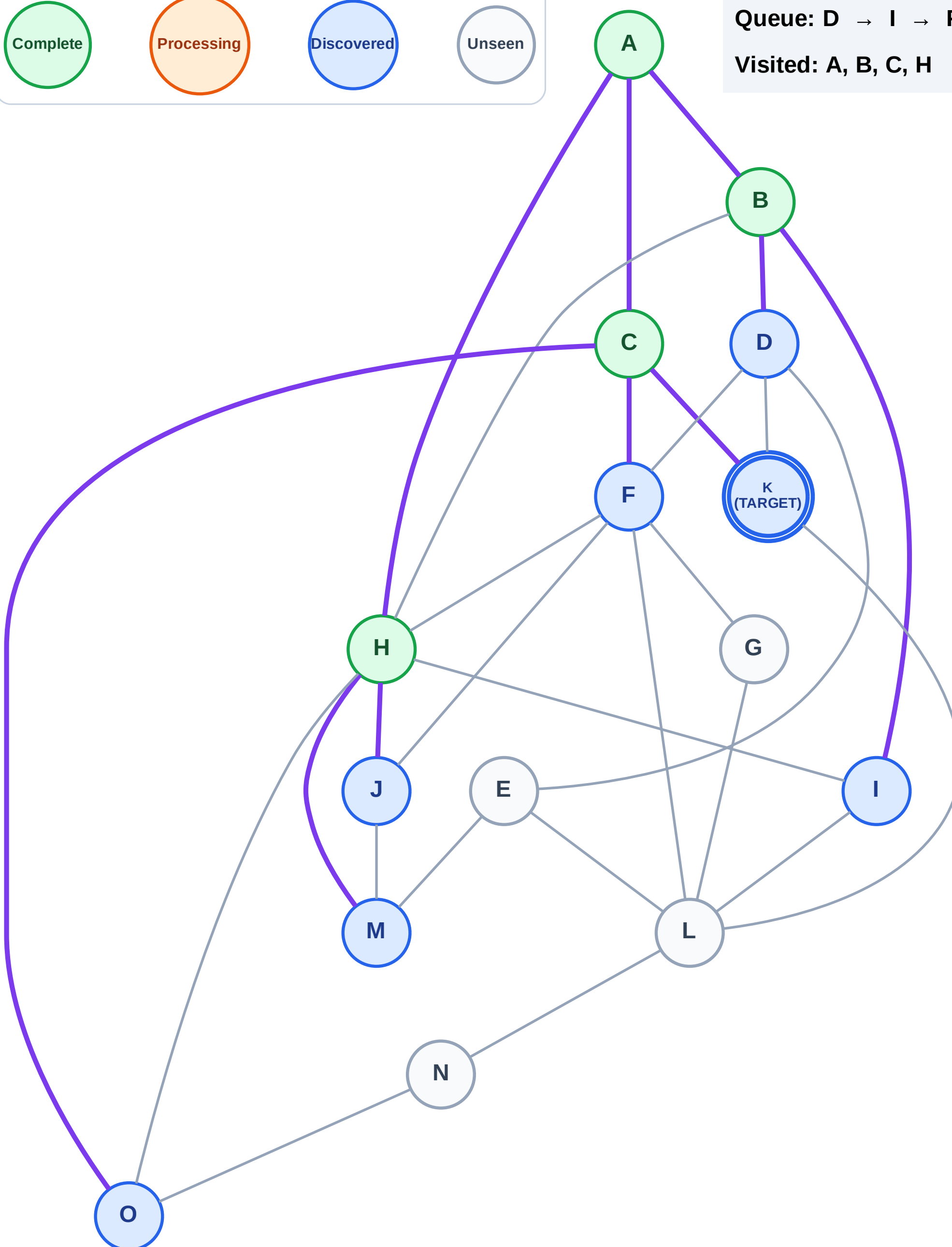
Step 9: Discover J, M from H

Legend



Queue: D → I → F → K → O → J → M

Visited: A, B, C, H



BFS Traversal

Step 10: Process node D

Legend

Complete

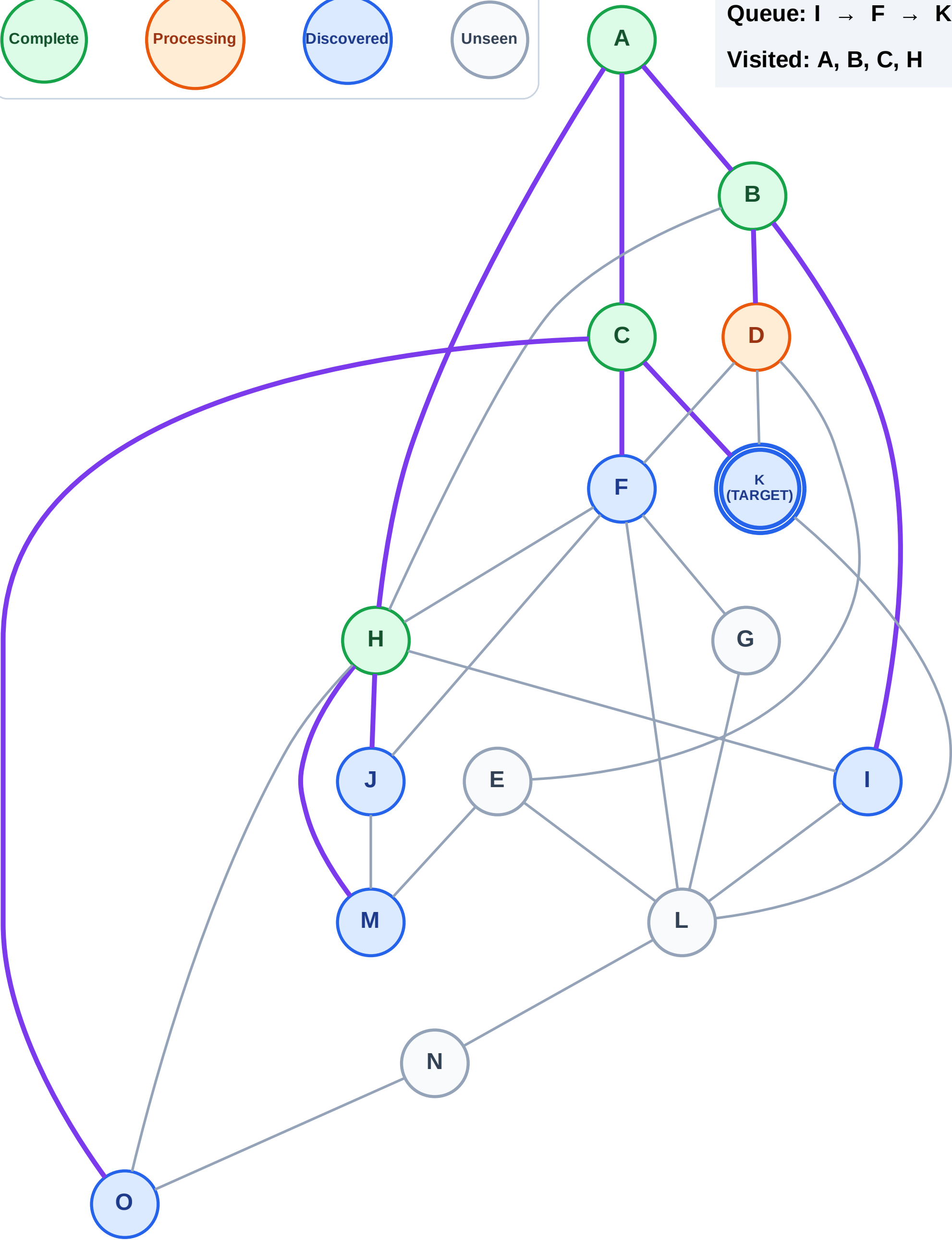
Processing

Discovered

Unseen

Queue: I → F → K → O → J → M

Visited: A, B, C, H



BFS Traversal

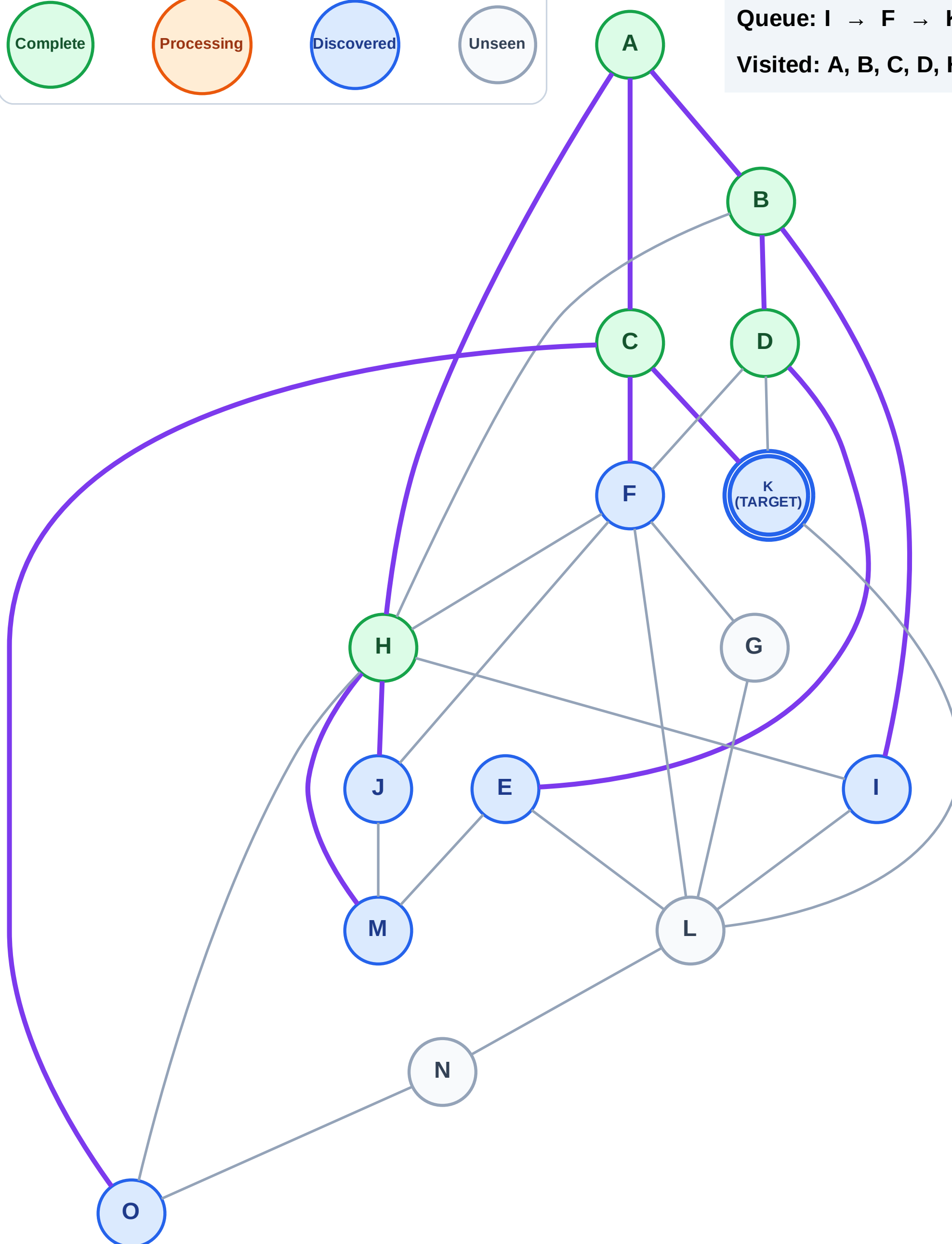
Step 11: Discover E from D

Legend



Queue: I → F → K → O → J → M → E

Visited: A, B, C, D, H



BFS Traversal

Step 12: Process node I

Legend

Complete

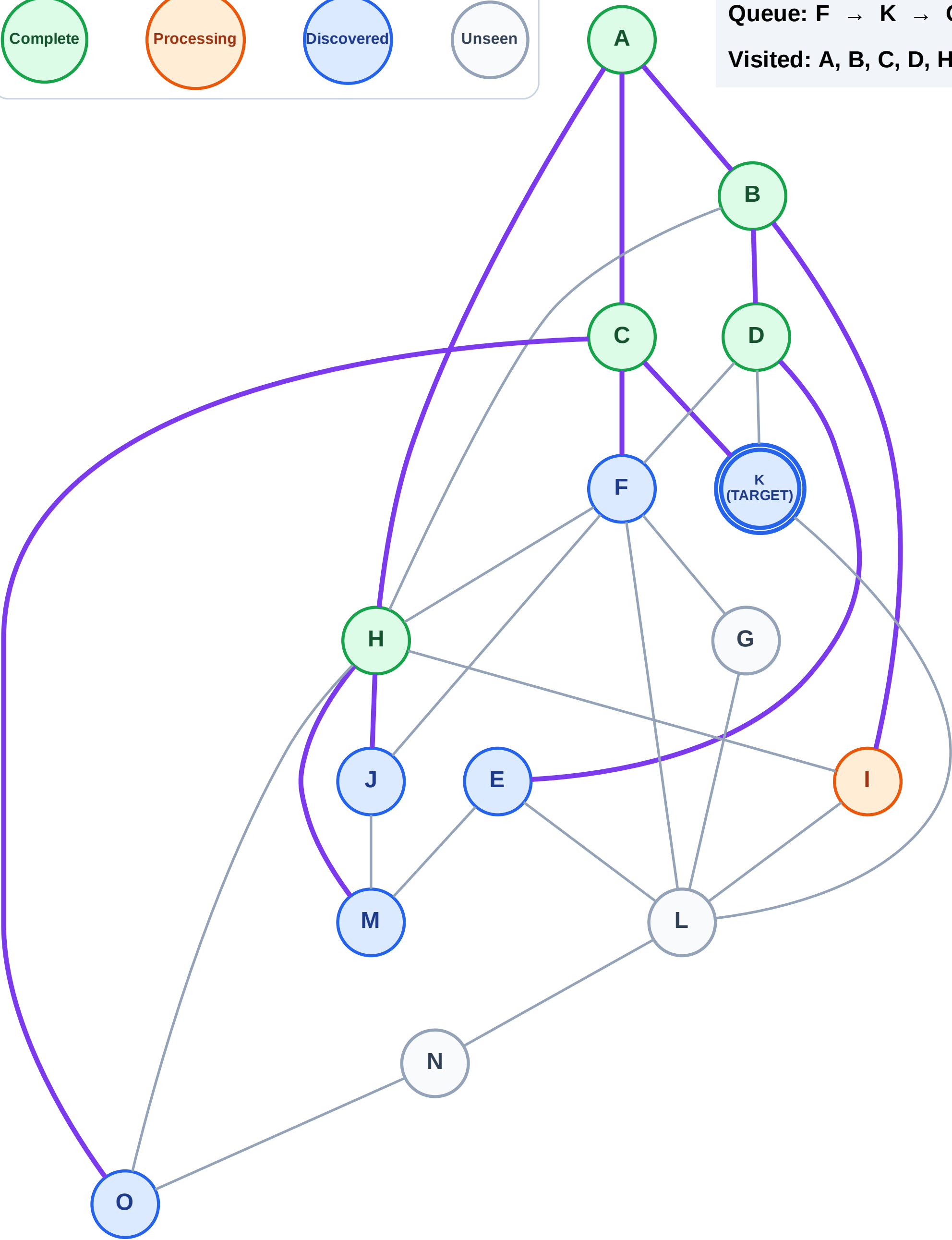
Processing

Discovered

Unseen

Queue: F → K → O → J → M → E

Visited: A, B, C, D, H



BFS Traversal

Step 13: Discover L from I

Legend

Complete

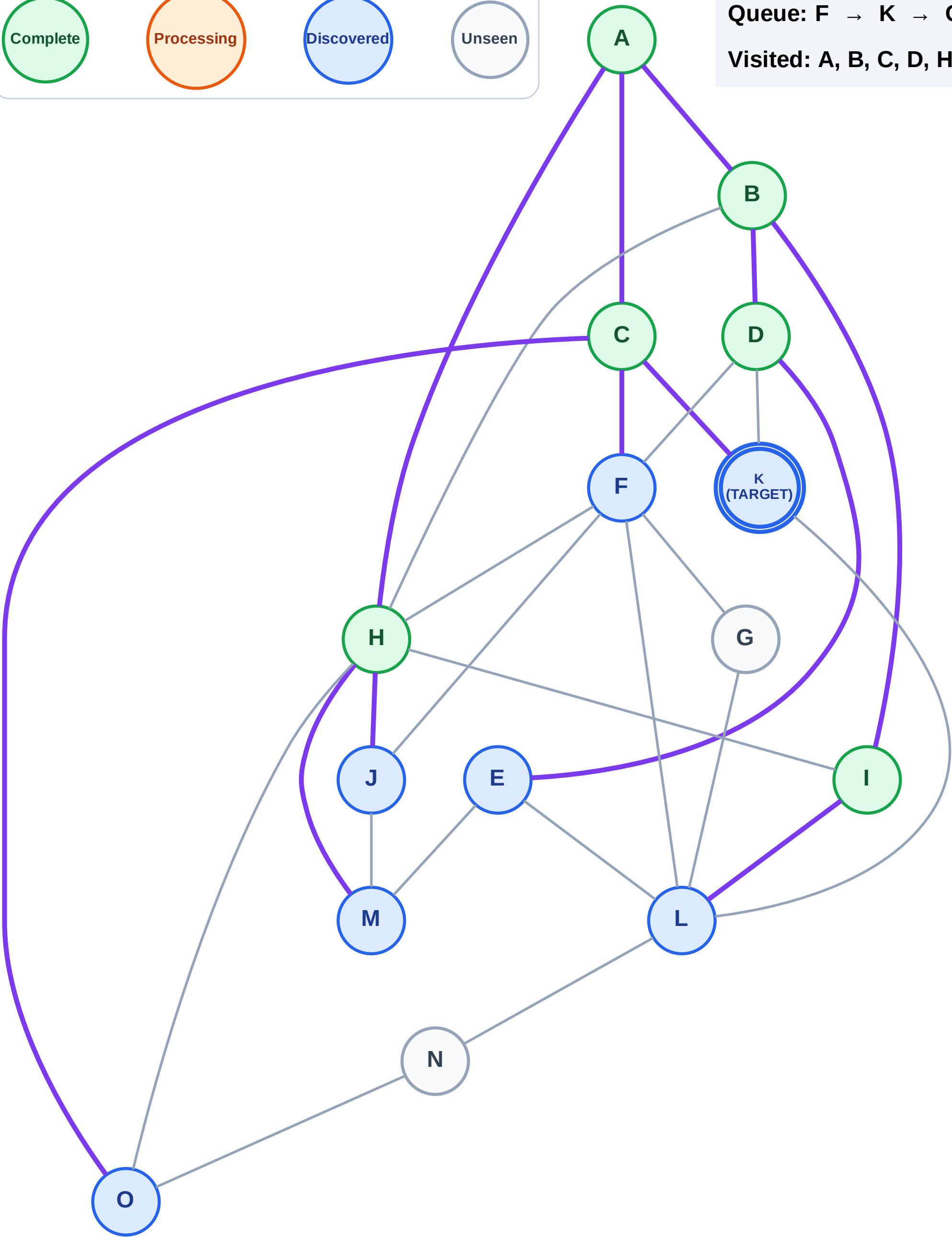
Processing

Discovered

Unseen

Queue: F → K → O → J → M → E → L

Visited: A, B, C, D, H, I



BFS Traversal

Step 14: Process node F

Legend

Complete

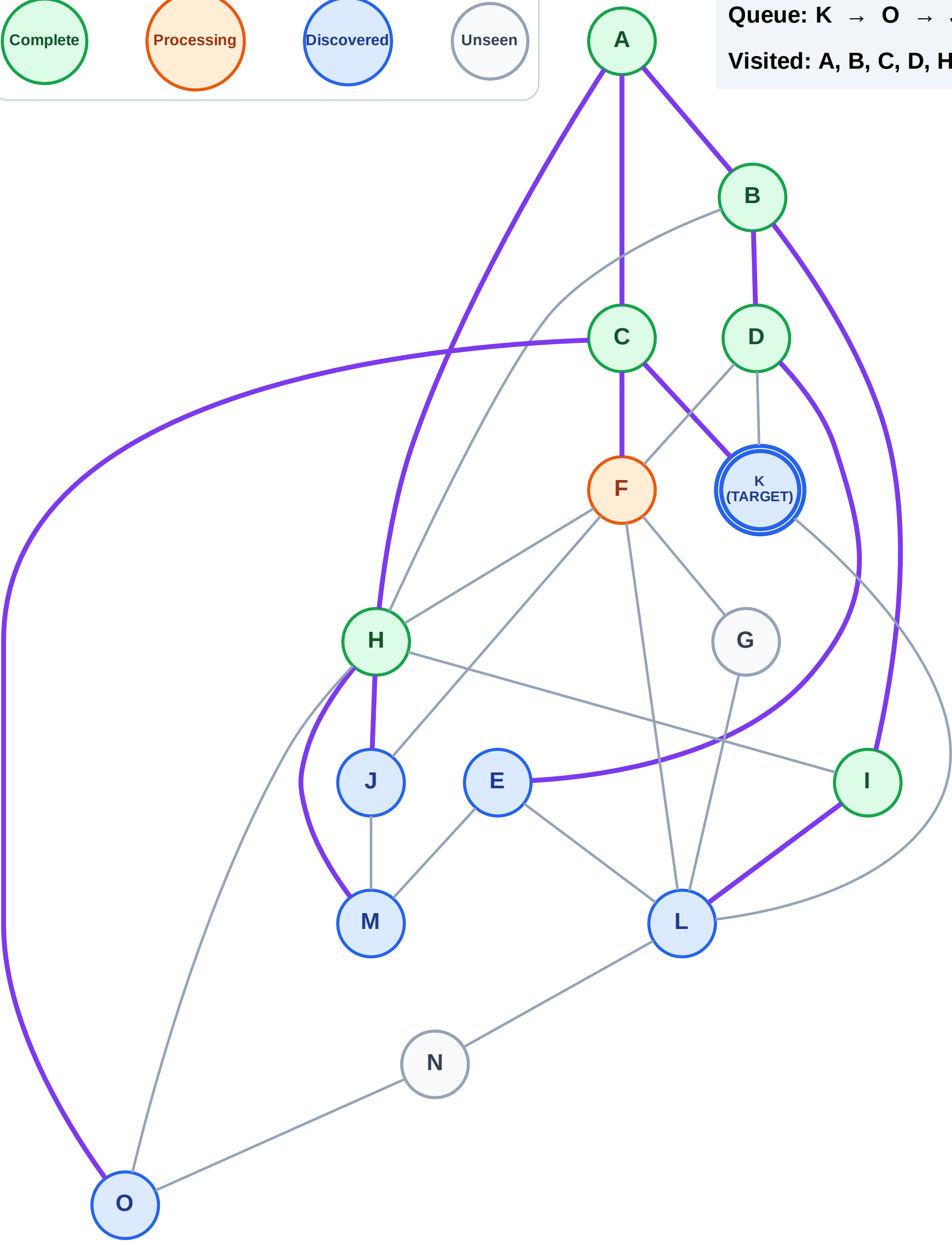
Processing

Discovered

Unseen

Queue: K → O → J → M → E → L

Visited: A, B, C, D, H, I



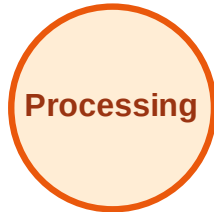
BFS Traversal

Step 15: Discover G from F

Legend



Complete



Processing



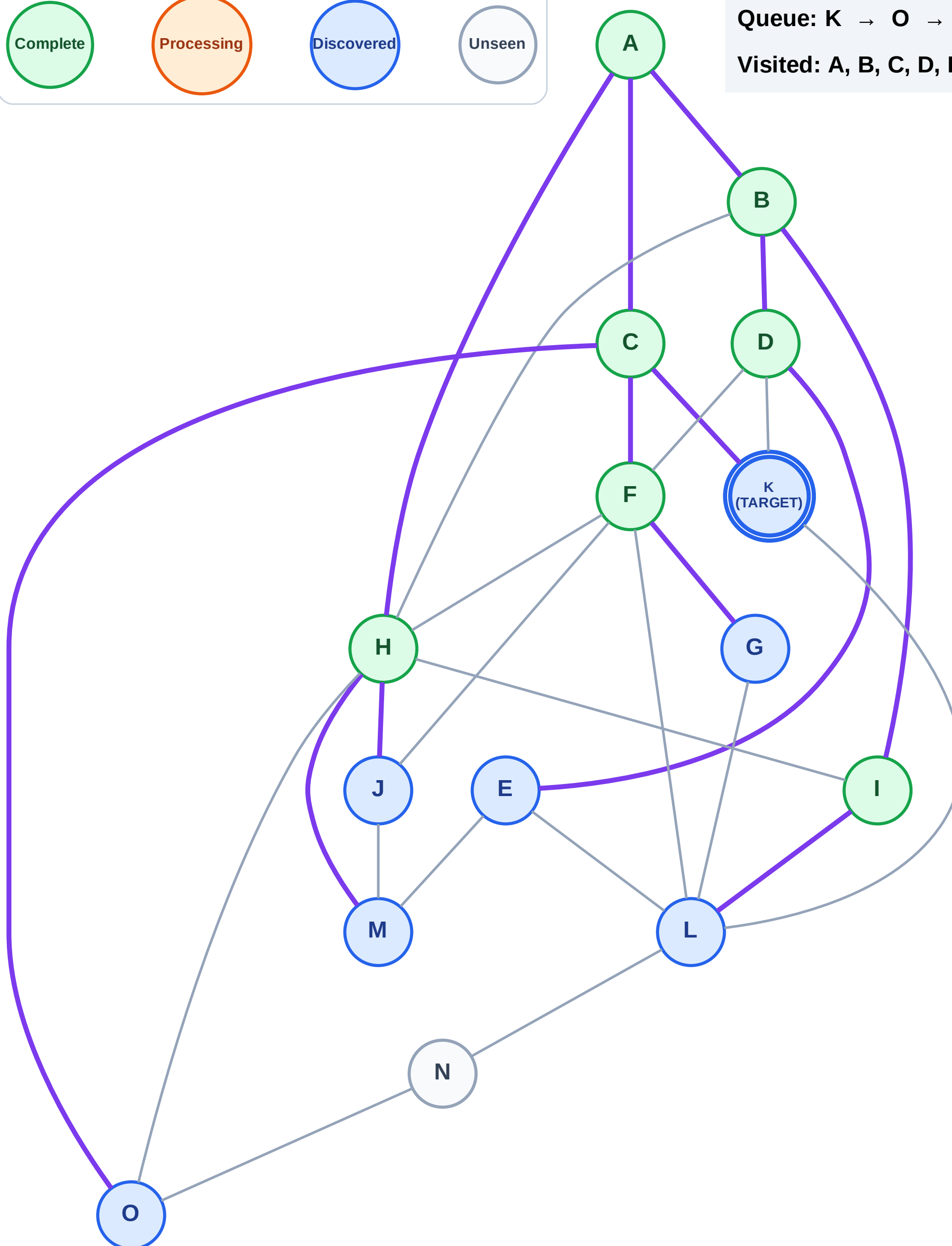
Discovered



Unseen

Queue: K → O → J → M → E → L → G

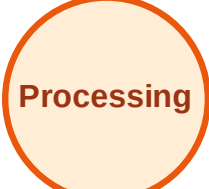
Visited: A, B, C, D, F, H, I



BFS Traversal

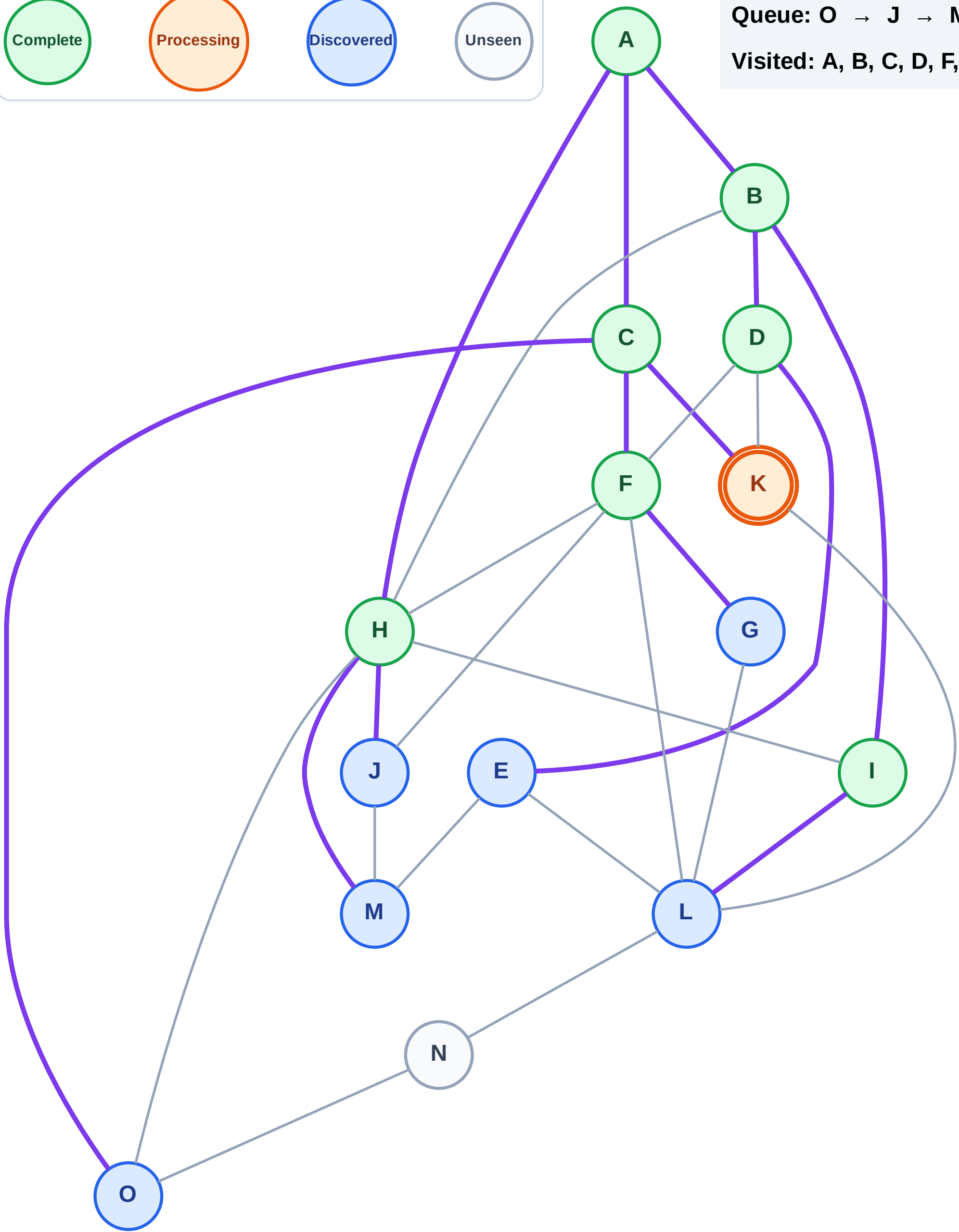
Step 16: Process node K

Legend



Queue: O → J → M → E → L → G

Visited: A, B, C, D, F, H, I



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

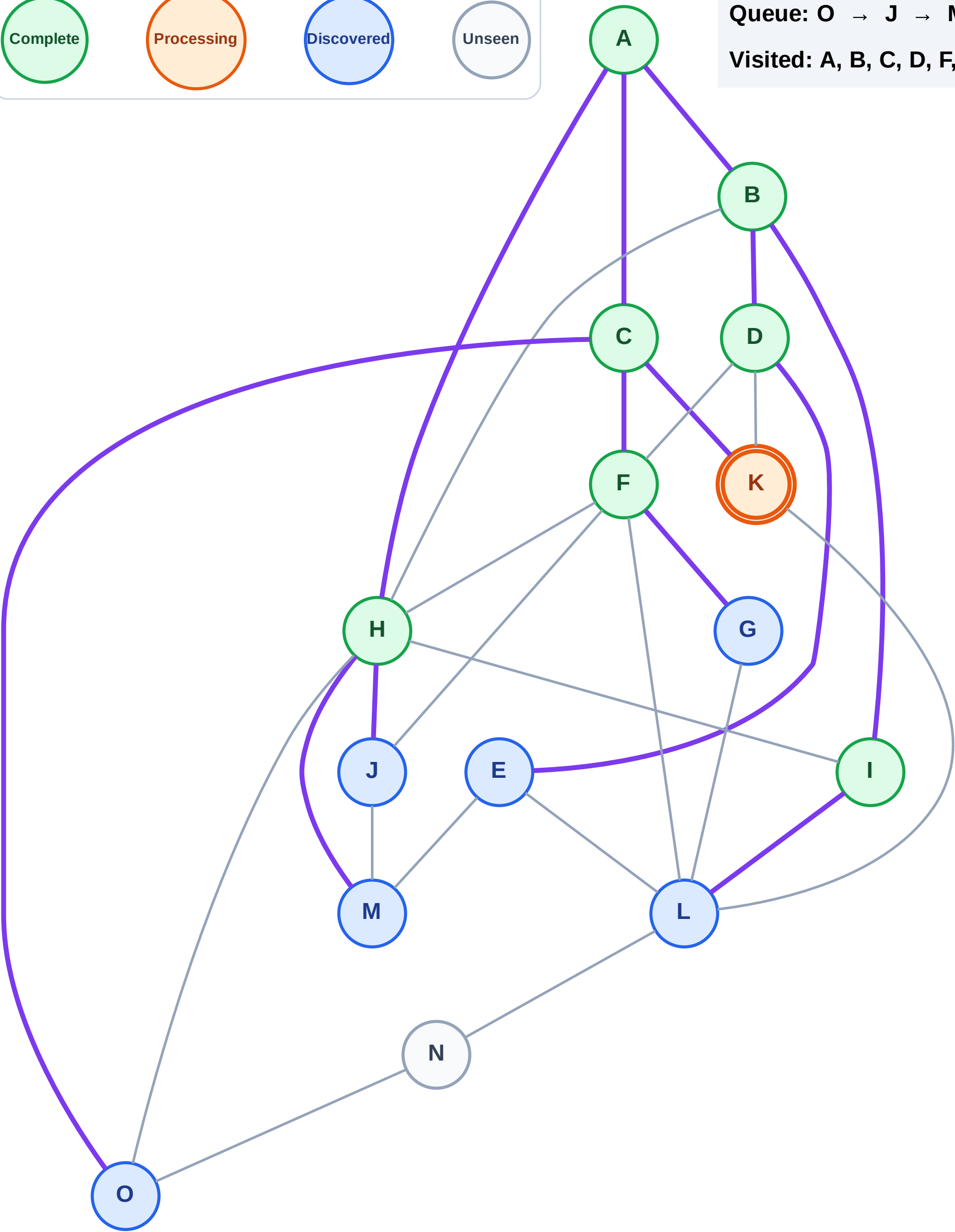
Processing

Discovered

Unseen

Queue: O → J → M → E → L → G

Visited: A, B, C, D, F, H, I, K



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

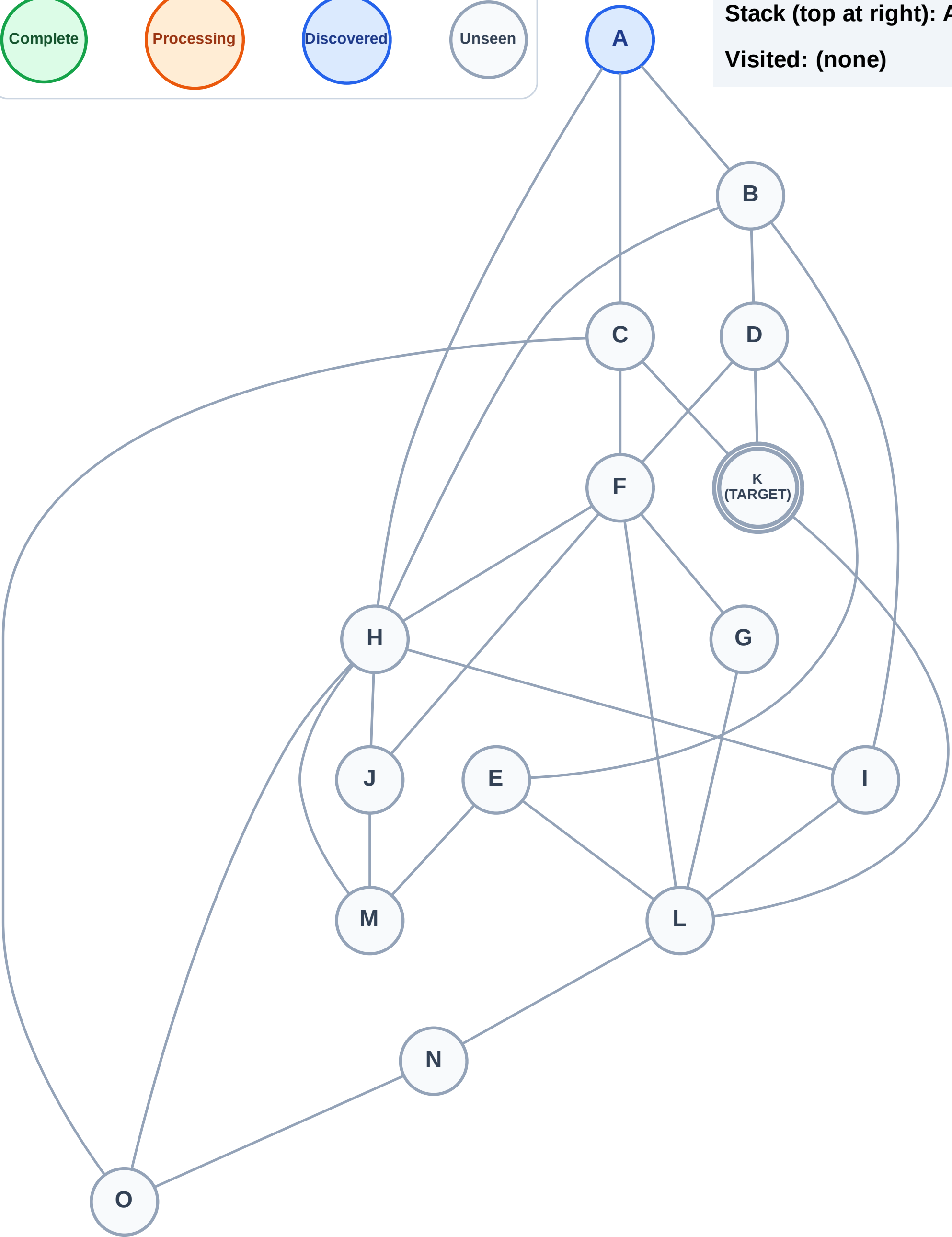
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

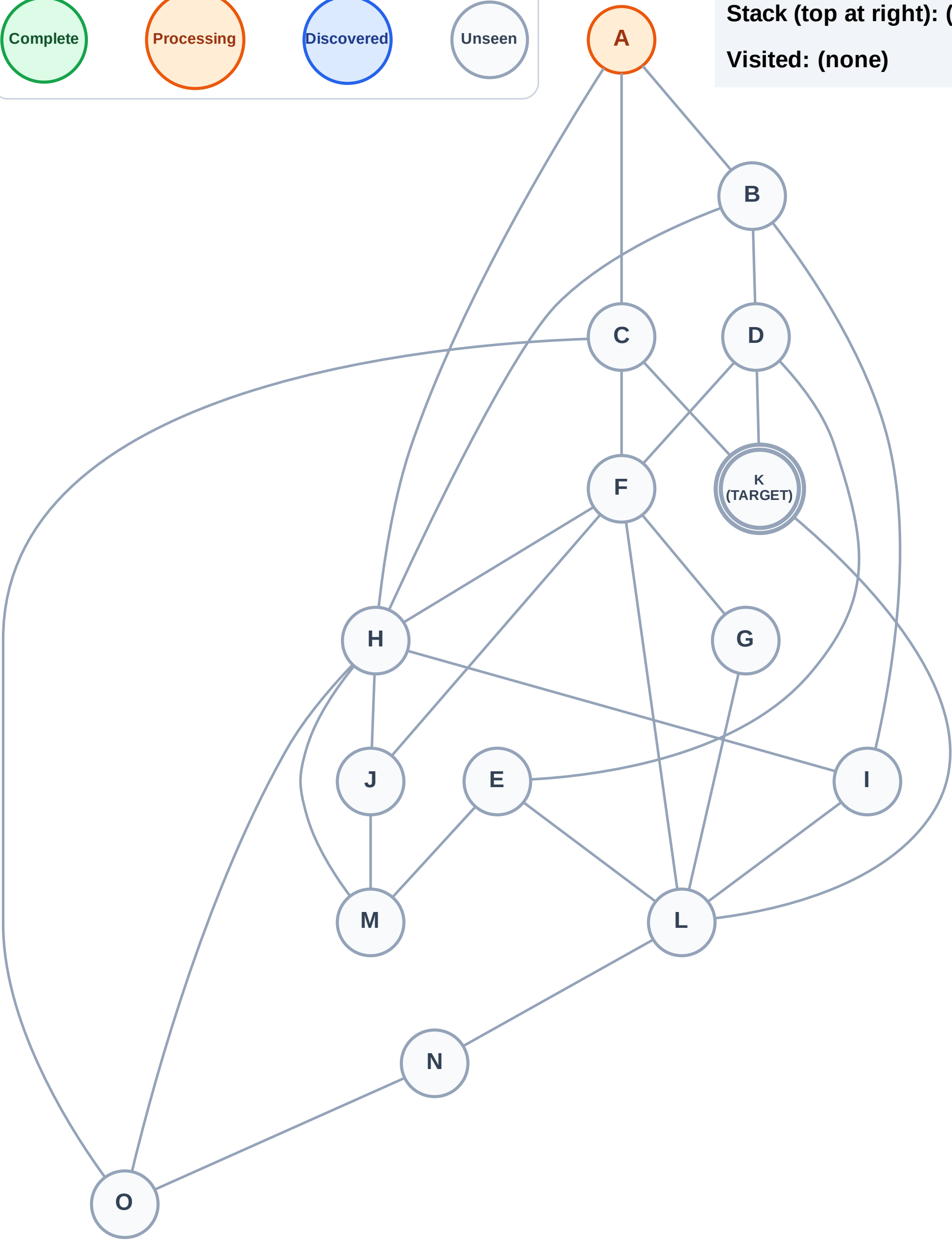
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

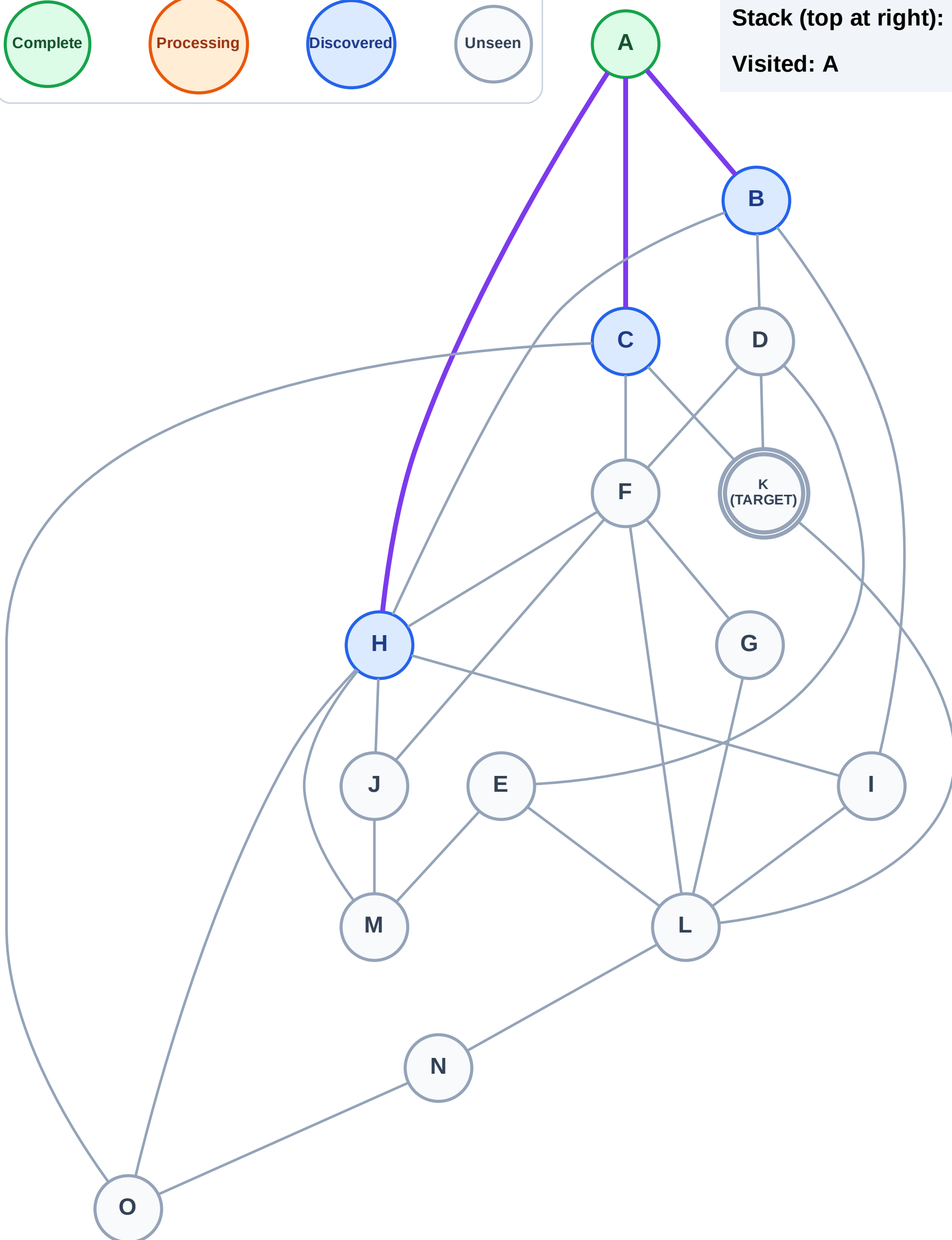
Step 3: Discover H, C, B from A

Legend



Stack (top at right): H | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

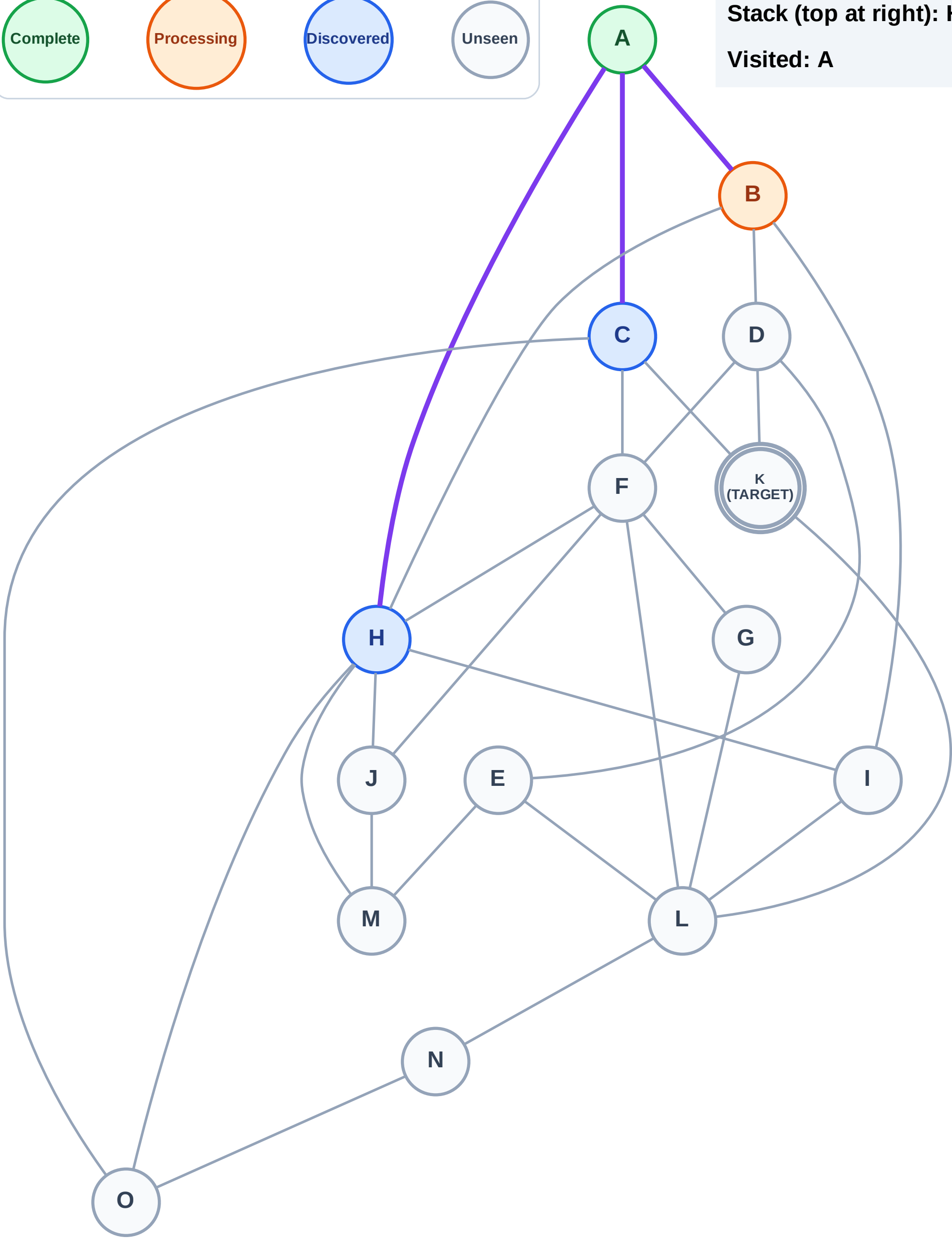
Processing

Discovered

Unseen

Stack (top at right): H | C

Visited: A



DFS Traversal

Step 5: Discover I, D from B

Legend

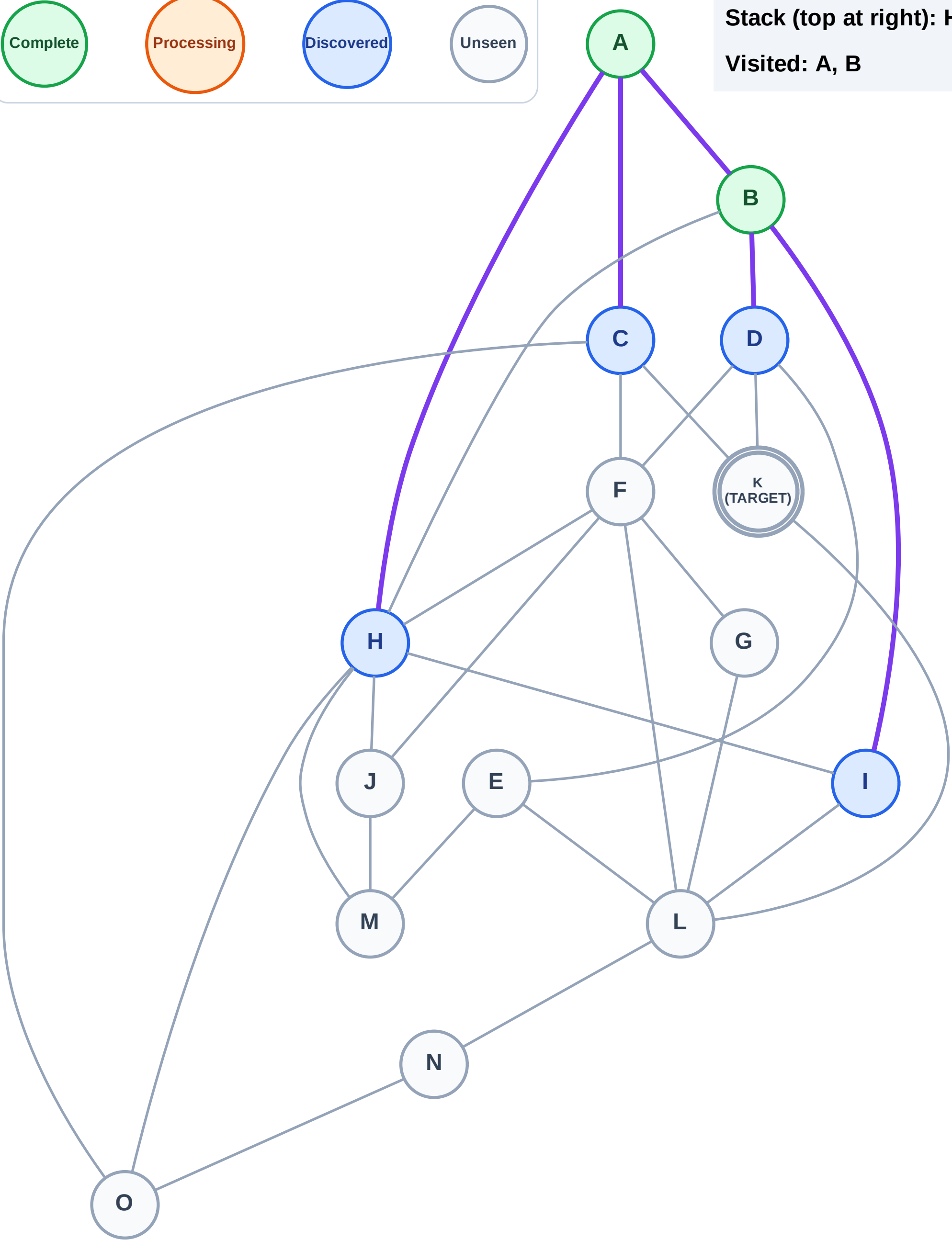
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I | D
Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

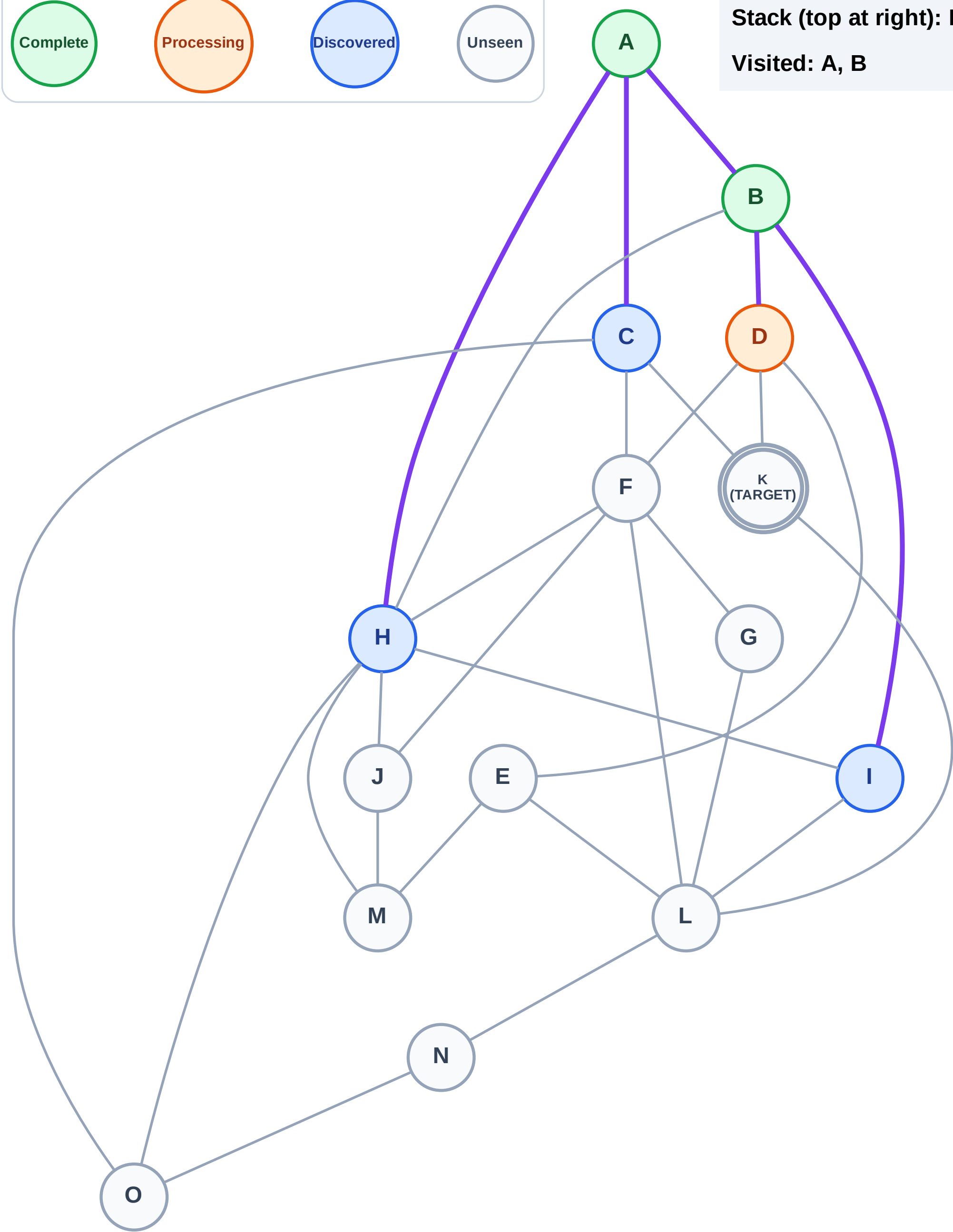
Processing

Discovered

Unseen

Stack (top at right): H | C | I

Visited: A, B



DFS Traversal

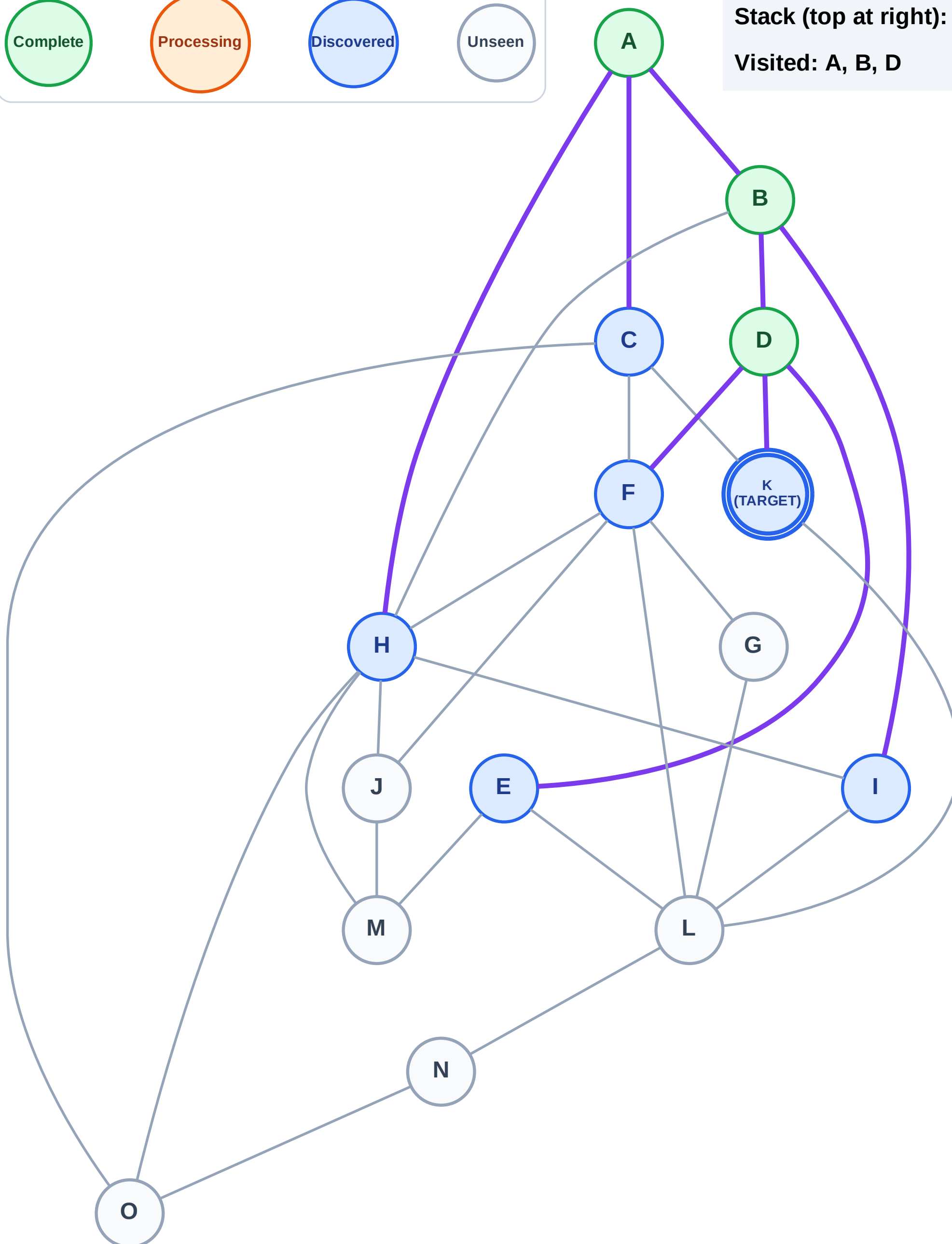
Step 7: Discover K, F, E from D

Legend



Stack (top at right): H | C | I | K | F | E

Visited: A, B, D



DFS Traversal

Step 8: Process node E

Legend

Complete

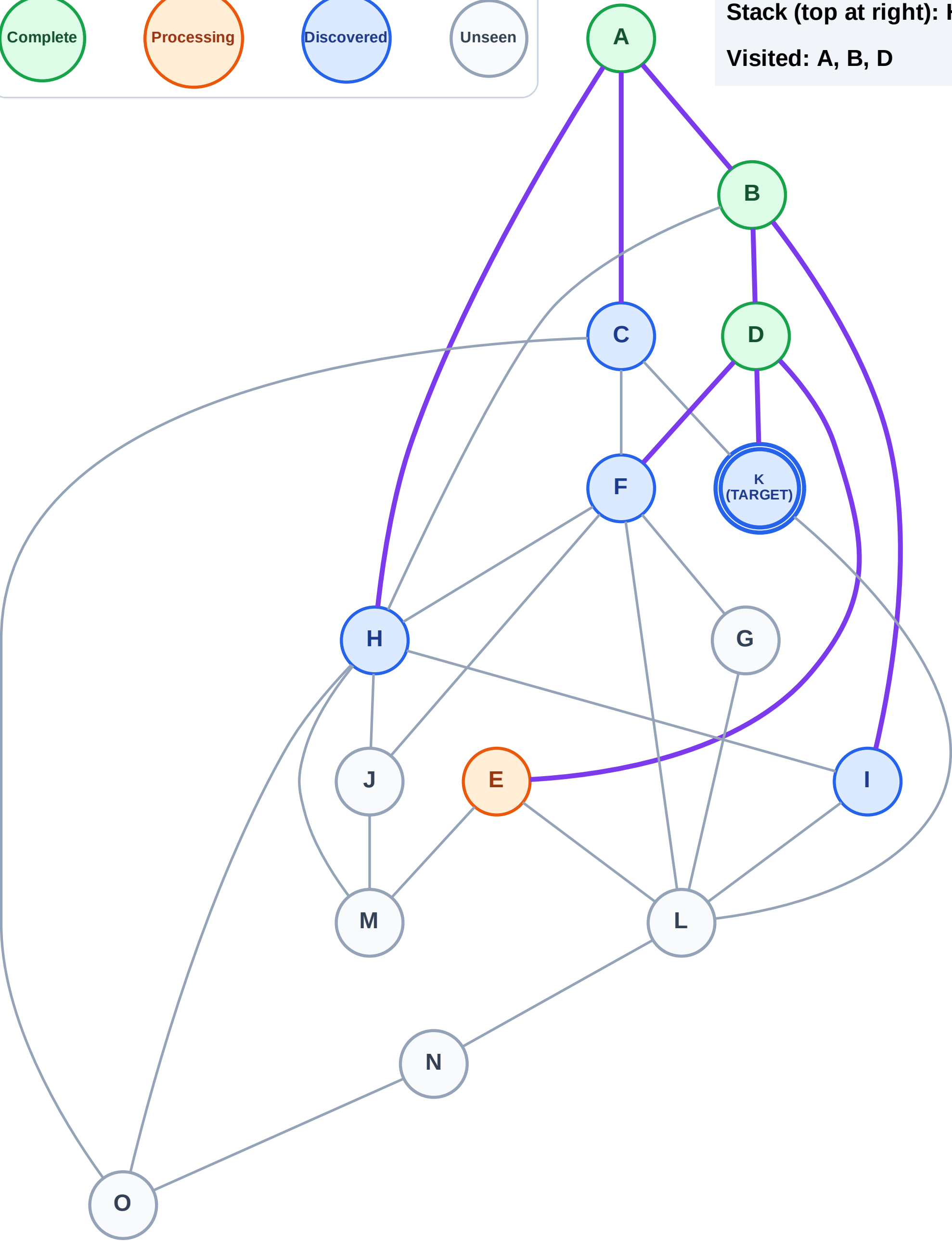
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F

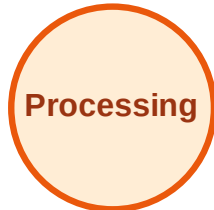
Visited: A, B, D



DFS Traversal

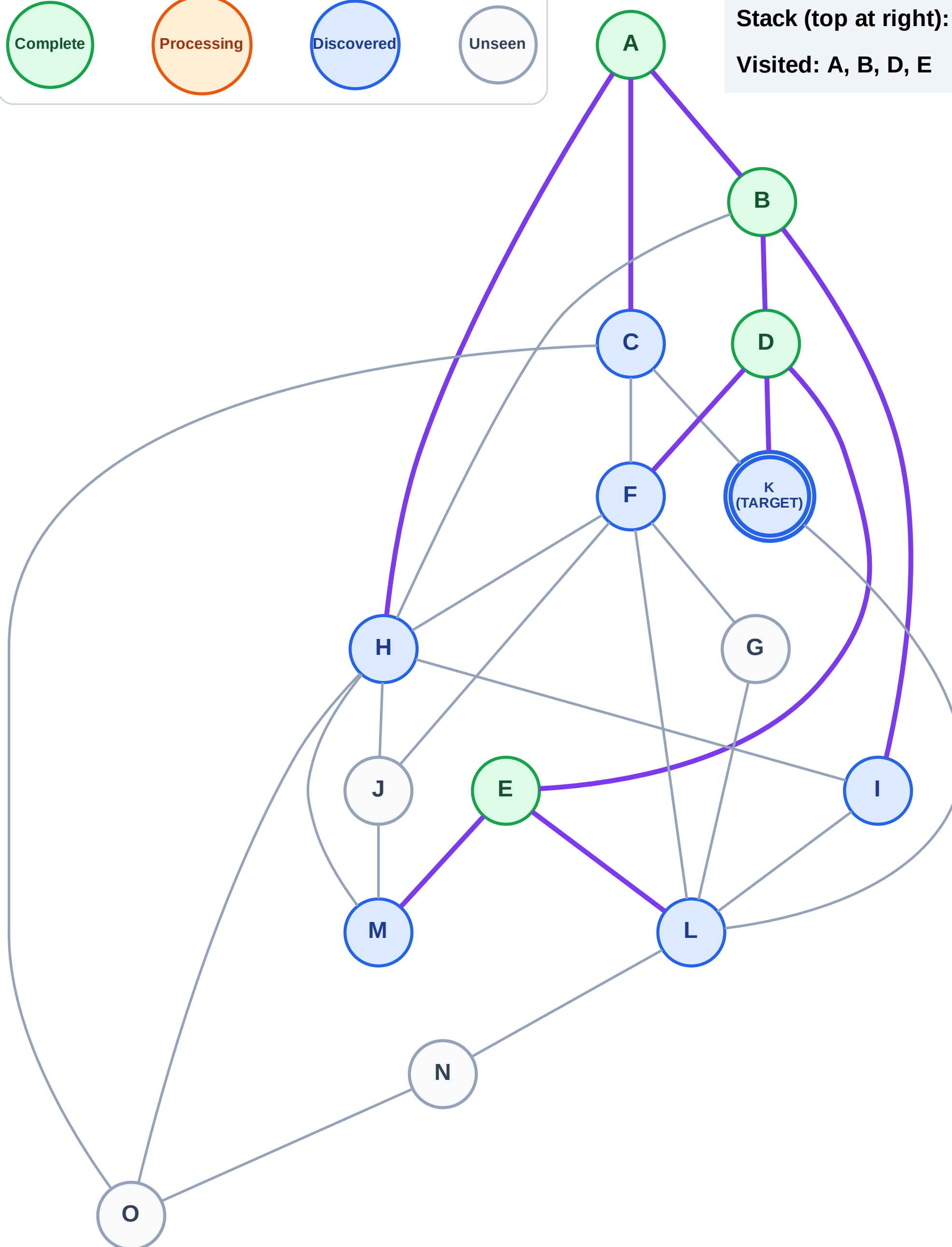
Step 9: Discover M, L from E

Legend



Stack (top at right): H | C | I | K | F | M | L

Visited: A, B, D, E



DFS Traversal

Step 10: Process node L

Legend

Complete

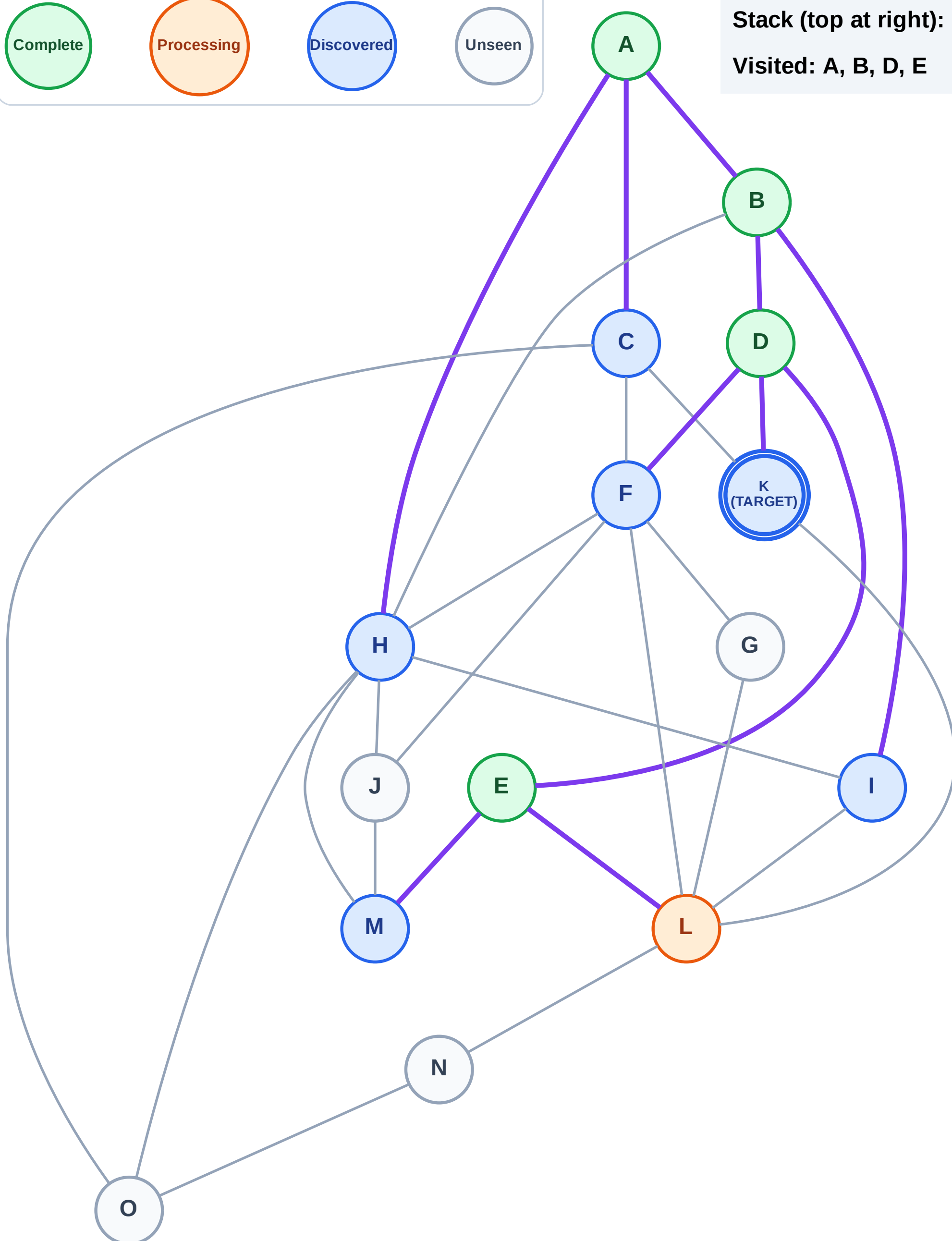
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F | M

Visited: A, B, D, E



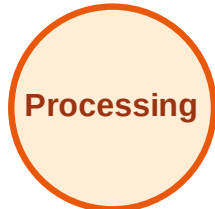
DFS Traversal

Step 11: Discover N, G from L

Legend



Complete



Processing



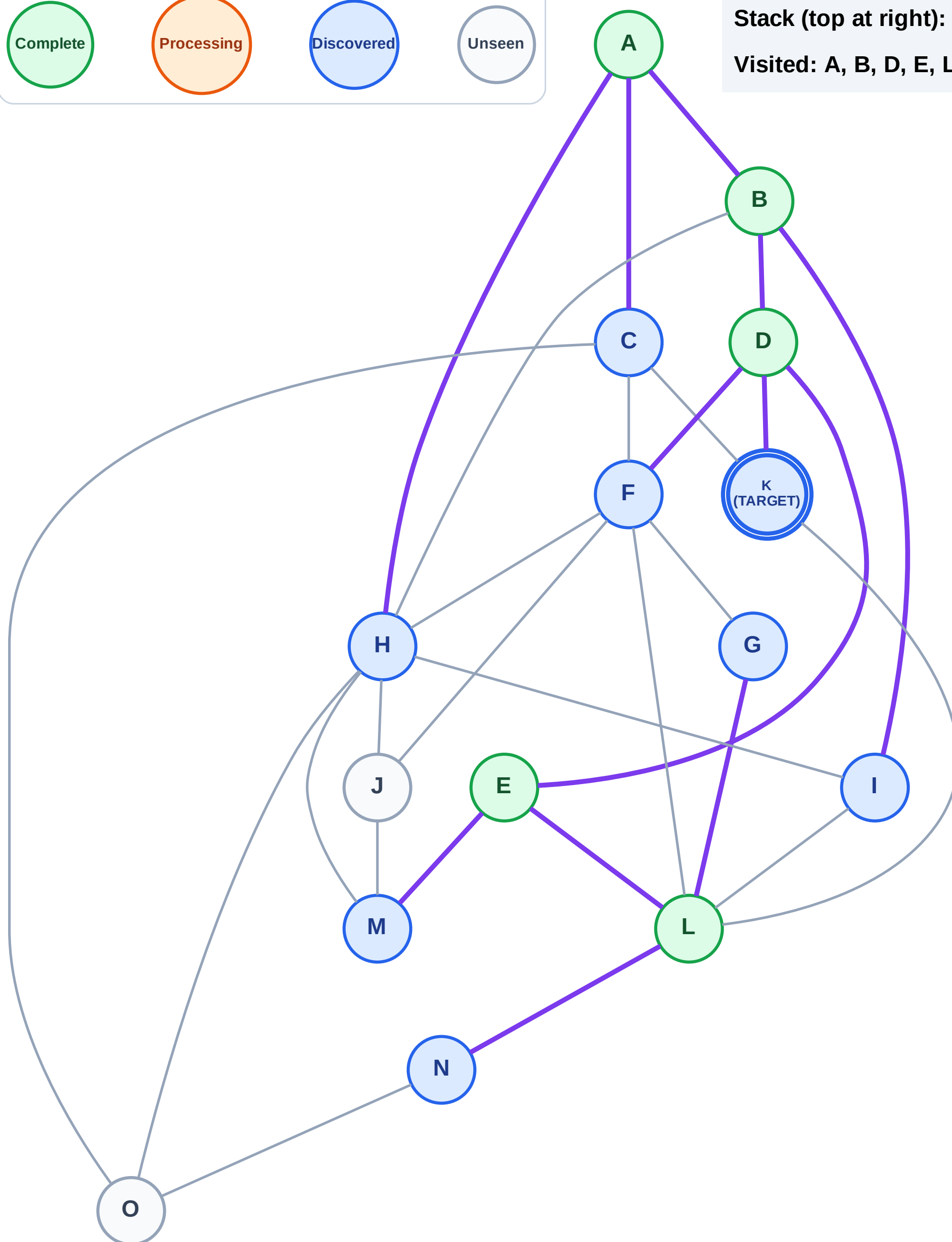
Discovered



Unseen

Stack (top at right): H | C | I | K | F | M | N | G

Visited: A, B, D, E, L



Step 12: Process node G



DFS Traversal

Step 13: No new neighbors from G

Legend



Complete



Processing



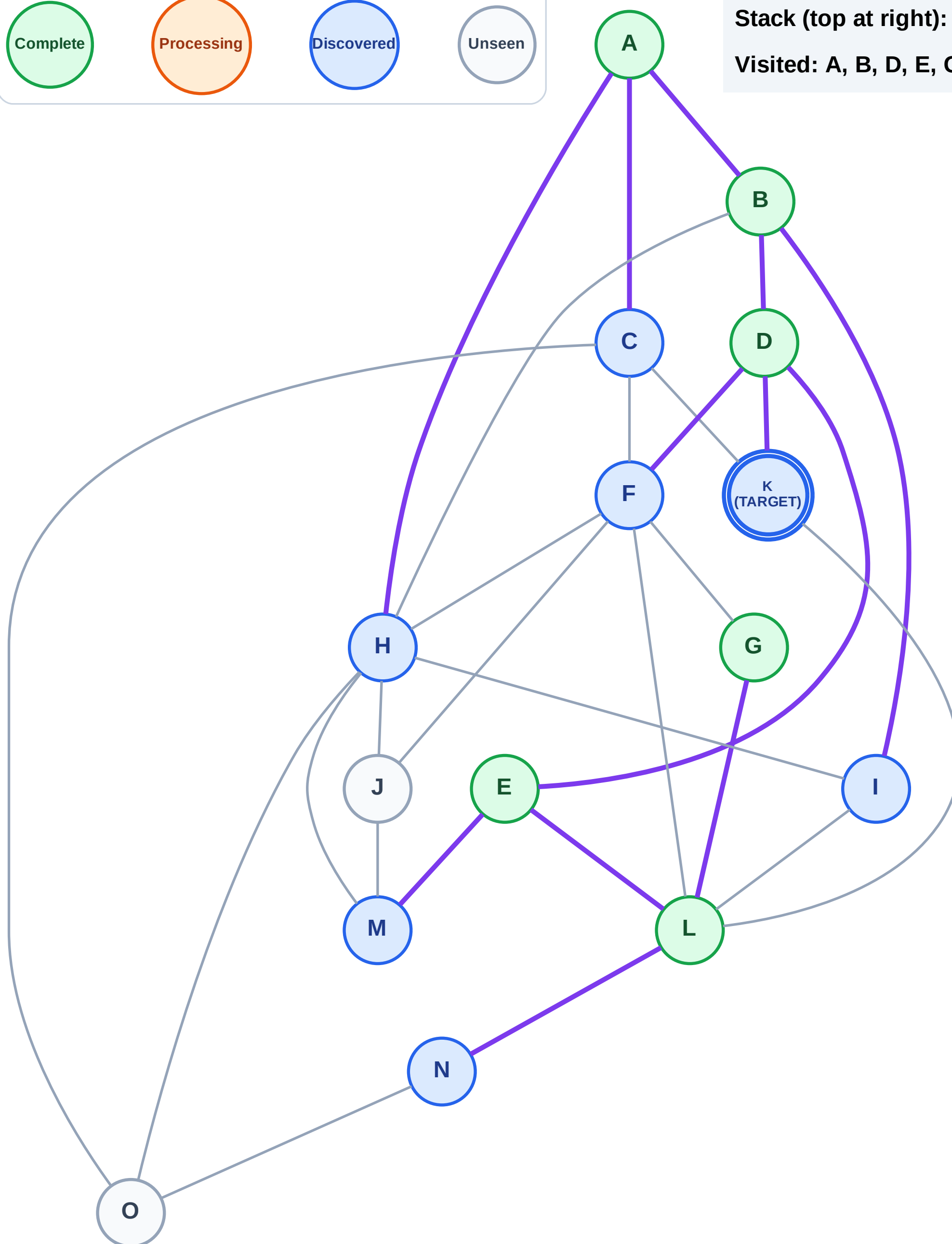
Discovered



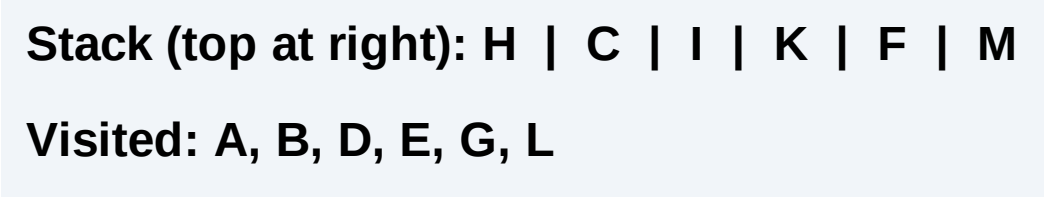
Unseen

Stack (top at right): H | C | I | K | F | M | N

Visited: A, B, D, E, G, L



Step 14: Process node N



DFS Traversal

Step 15: Discover O from N

Legend



Complete



Processing



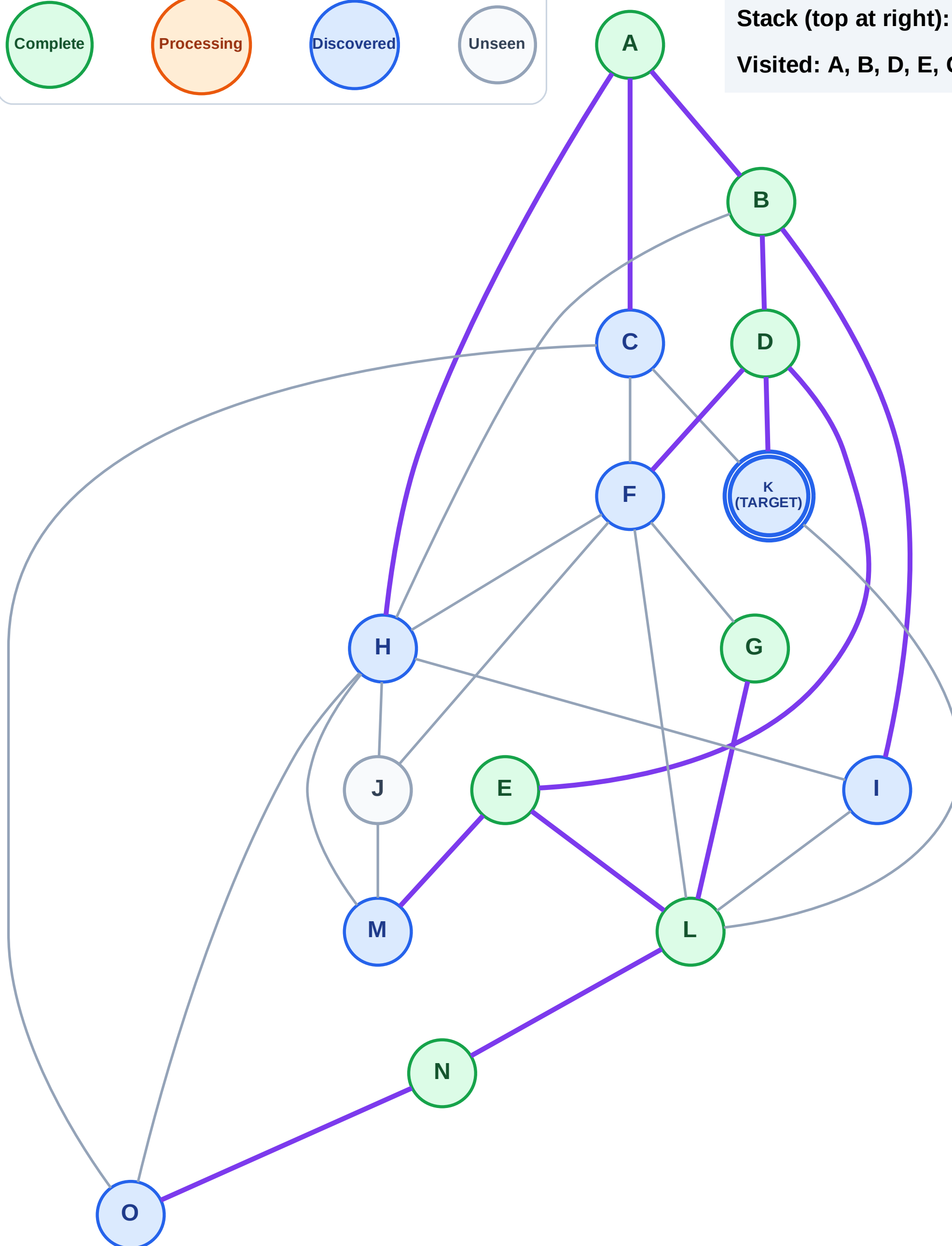
Discovered



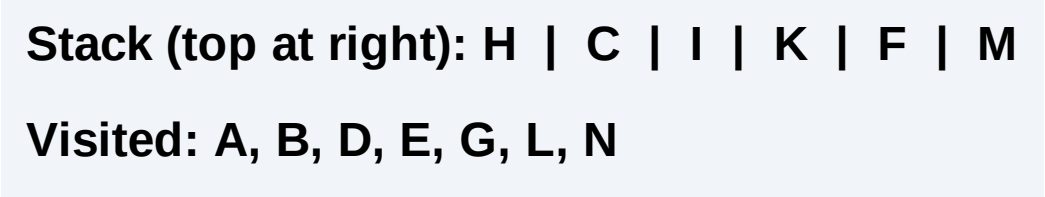
Unseen

Stack (top at right): H | C | I | K | F | M | O

Visited: A, B, D, E, G, L, N



Step 16: Process node O



DFS Traversal

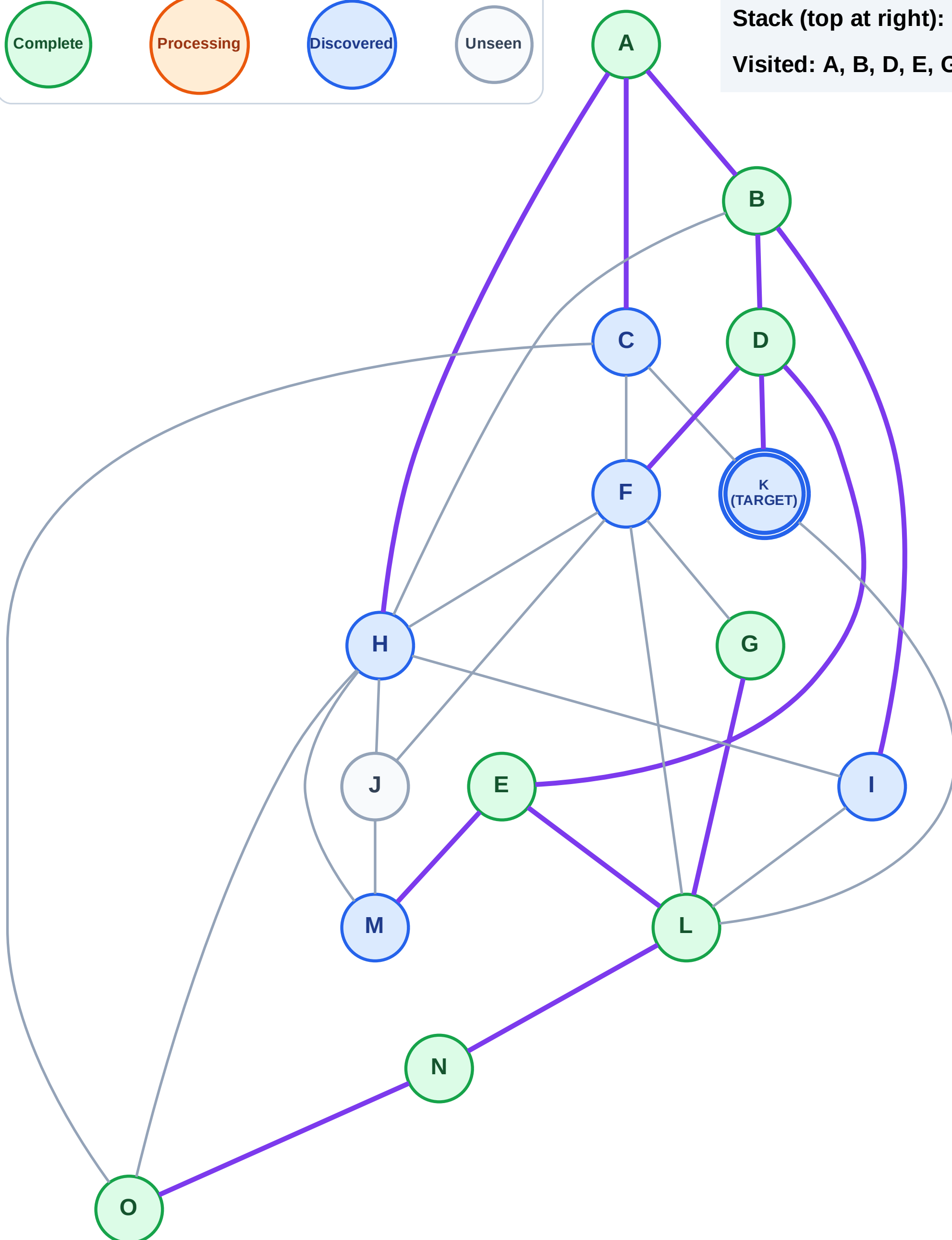
Step 17: No new neighbors from O

Legend



Stack (top at right): H | C | I | K | F | M

Visited: A, B, D, E, G, L, N, O



DFS Traversal

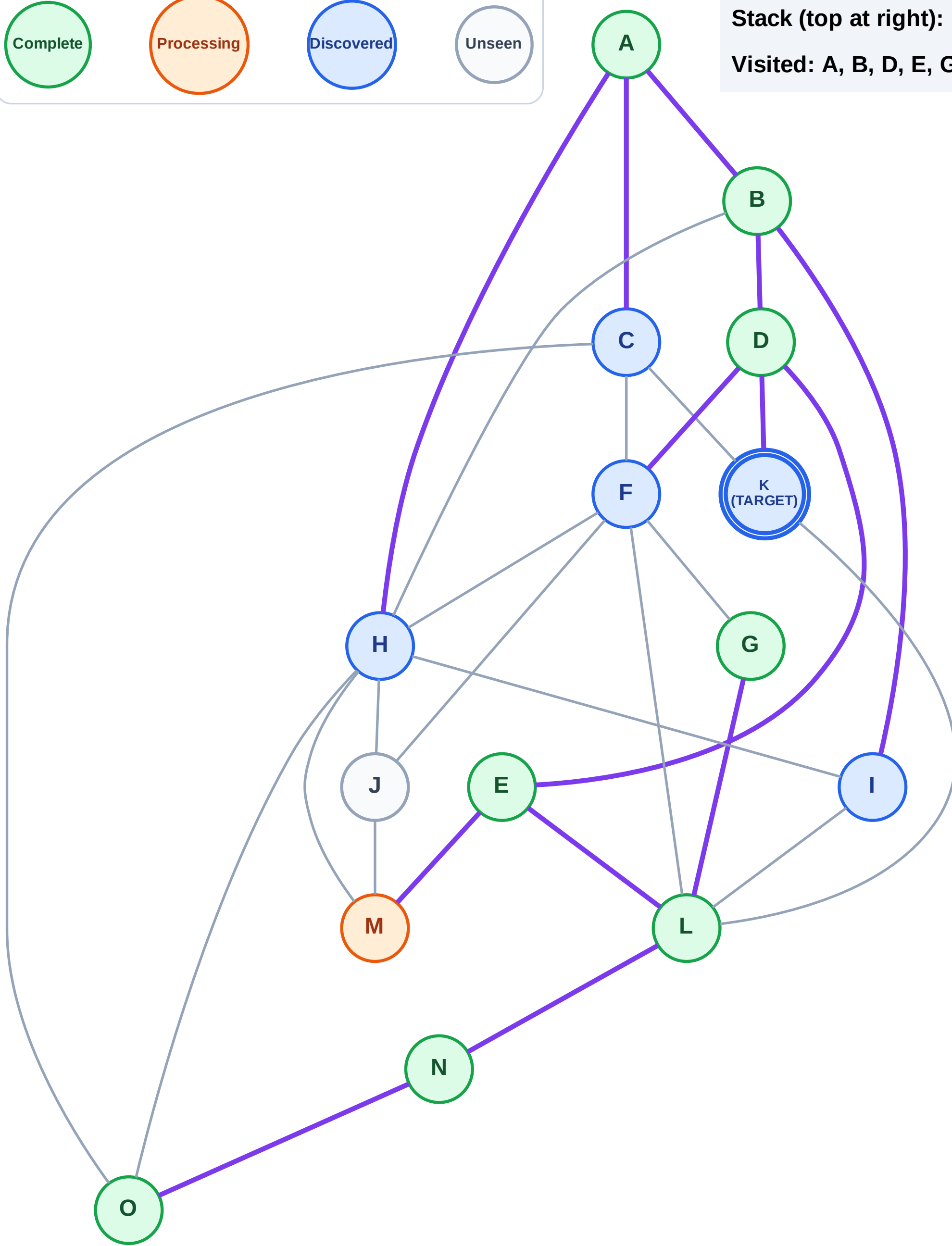
Step 18: Process node M

Legend



Stack (top at right): H | C | I | K | F

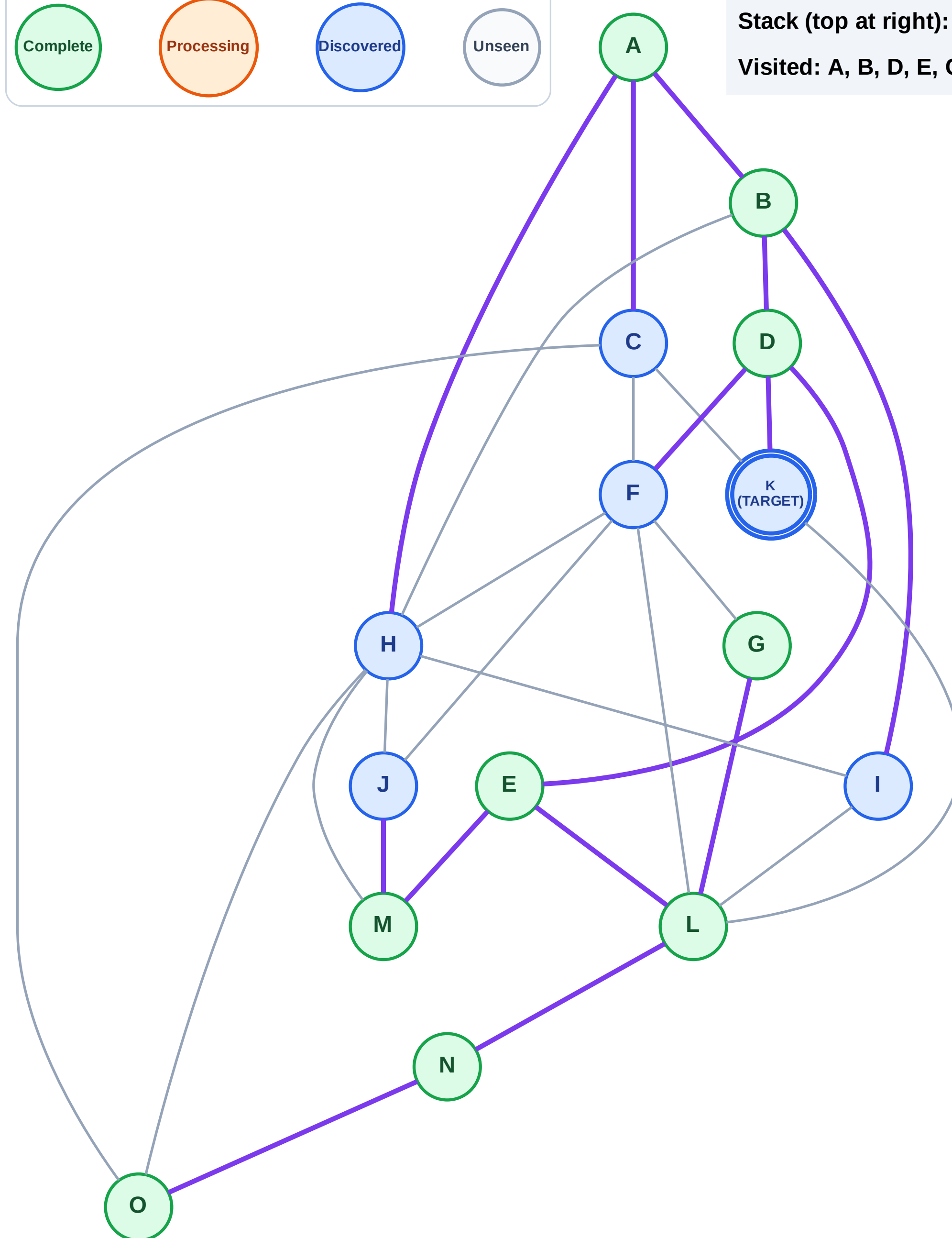
Visited: A, B, D, E, G, L, N, O



Step 19: Discover J from M

Stack (top at right): H | C | I | K | F | J

Visited: A, B, D, E, G, L, M, N, O



DFS Traversal

Step 20: Process node J

Stack (top at right): H | C | I | K | F
Visited: A, B, D, E, G, L, M, N, O

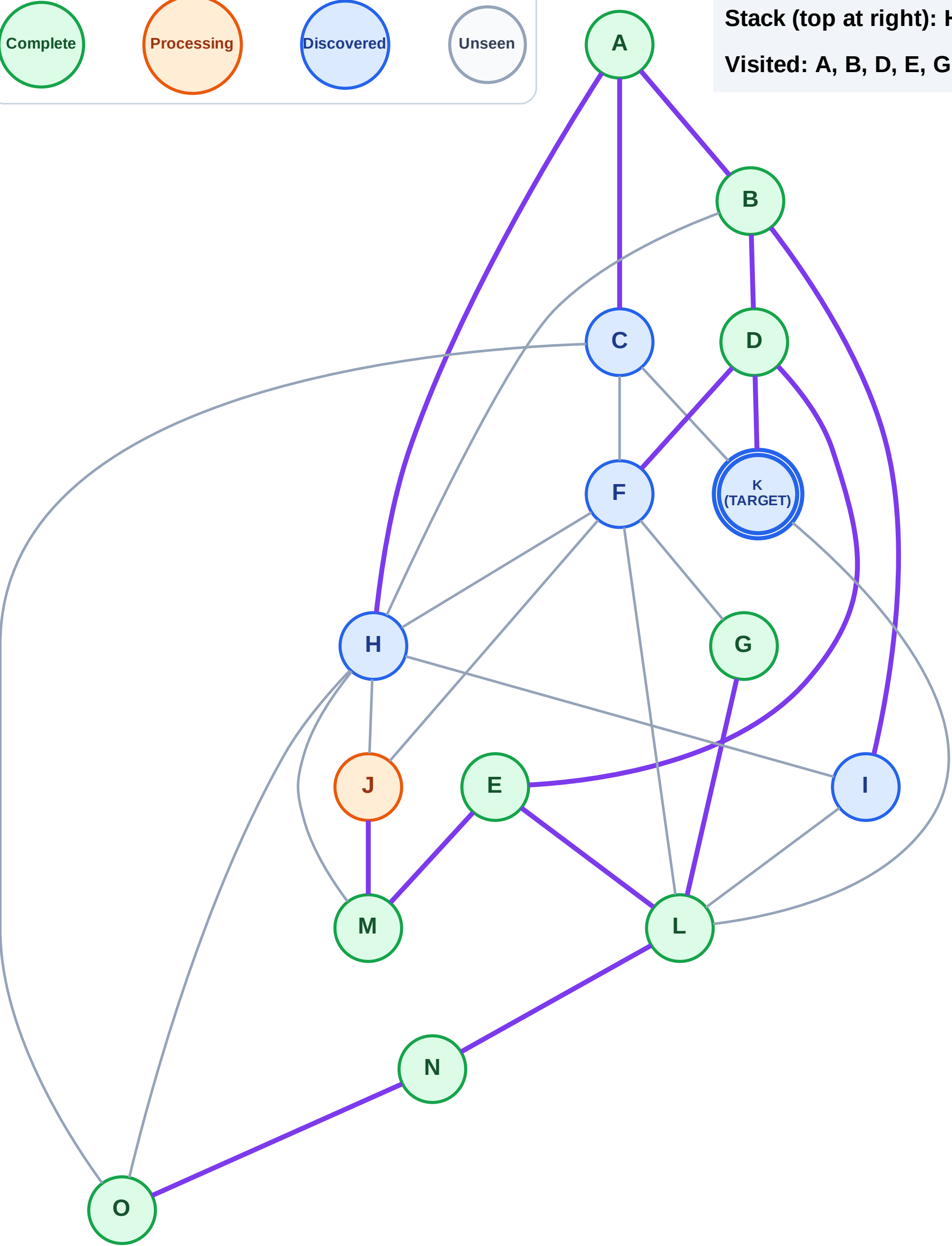
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

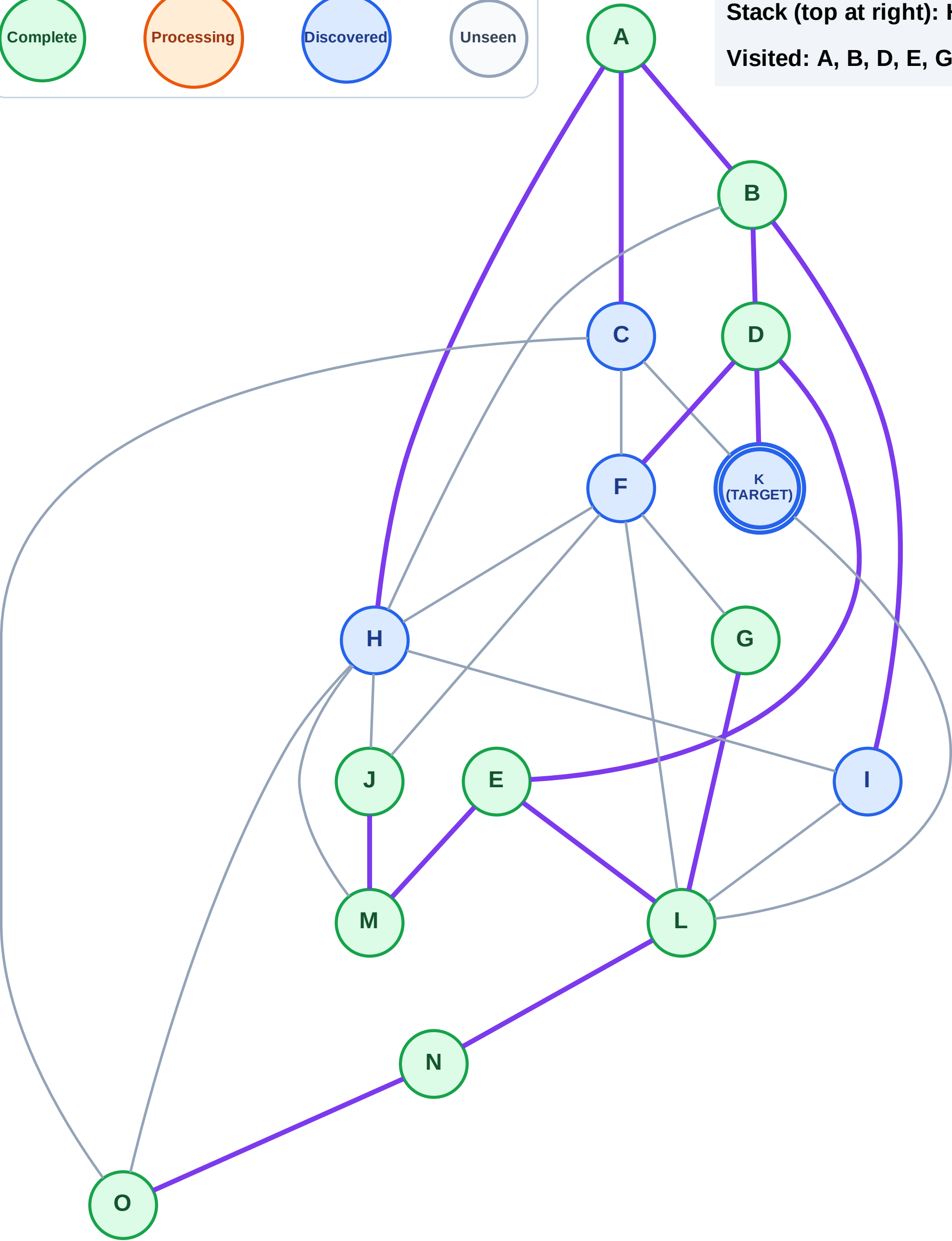
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F

Visited: A, B, D, E, G, J, L, M, N, O



DFS Traversal

Step 22: Process node F

Legend

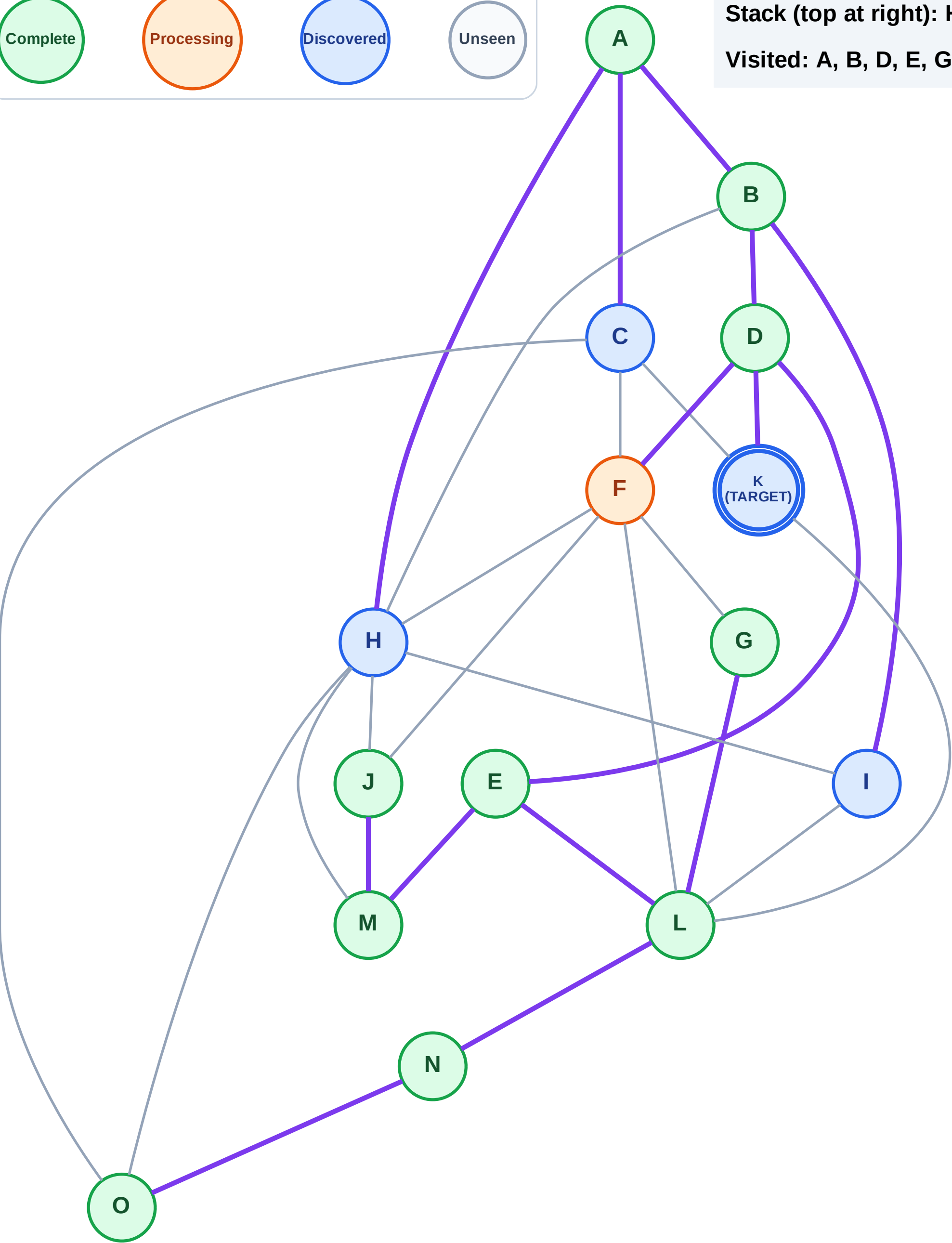
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I | K
Visited: A, B, D, E, G, J, L, M, N, O



DFS Traversal

Step 23: No new neighbors from F

Legend

Complete

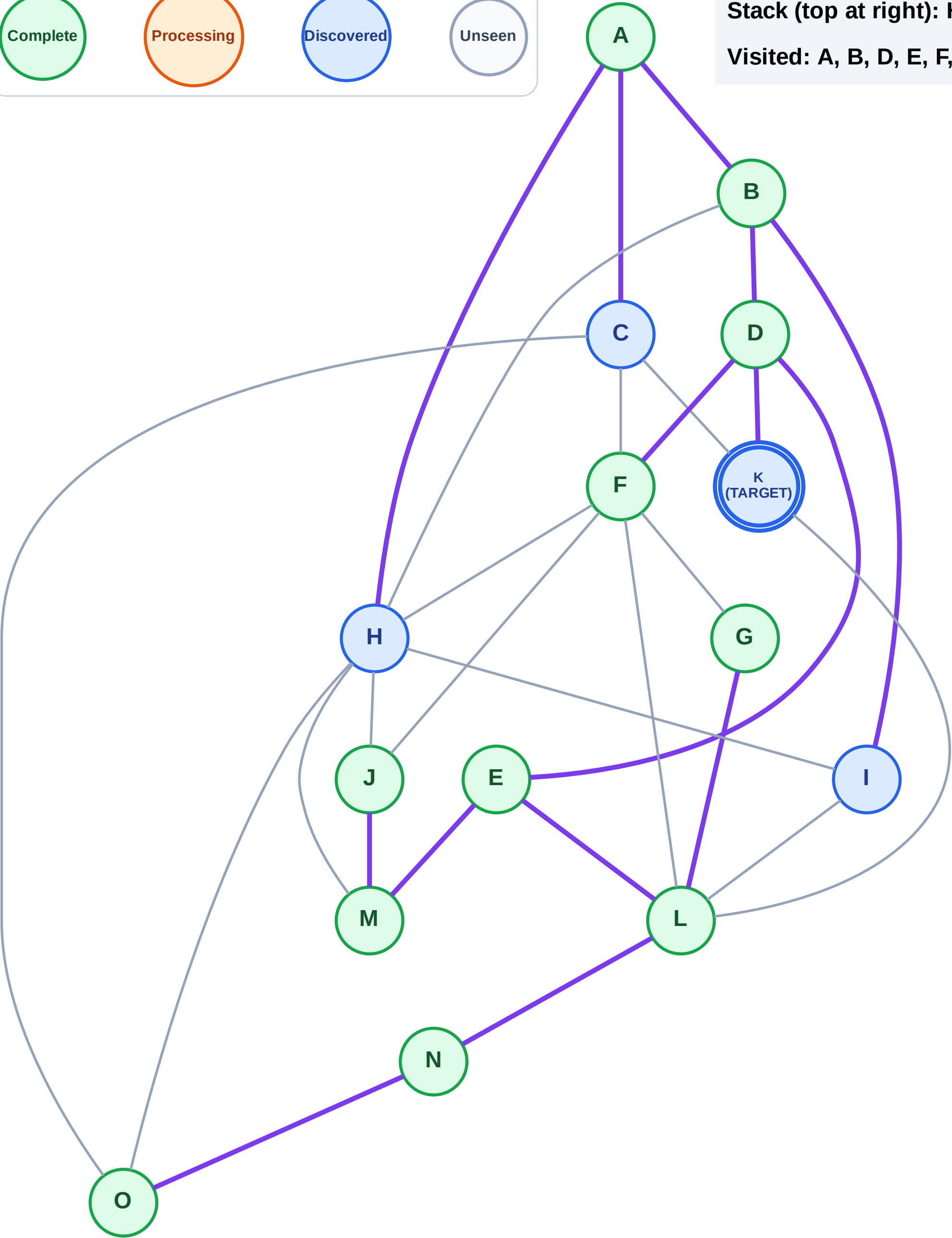
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K

Visited: A, B, D, E, F, G, J, L, M, N, O



DFS Traversal

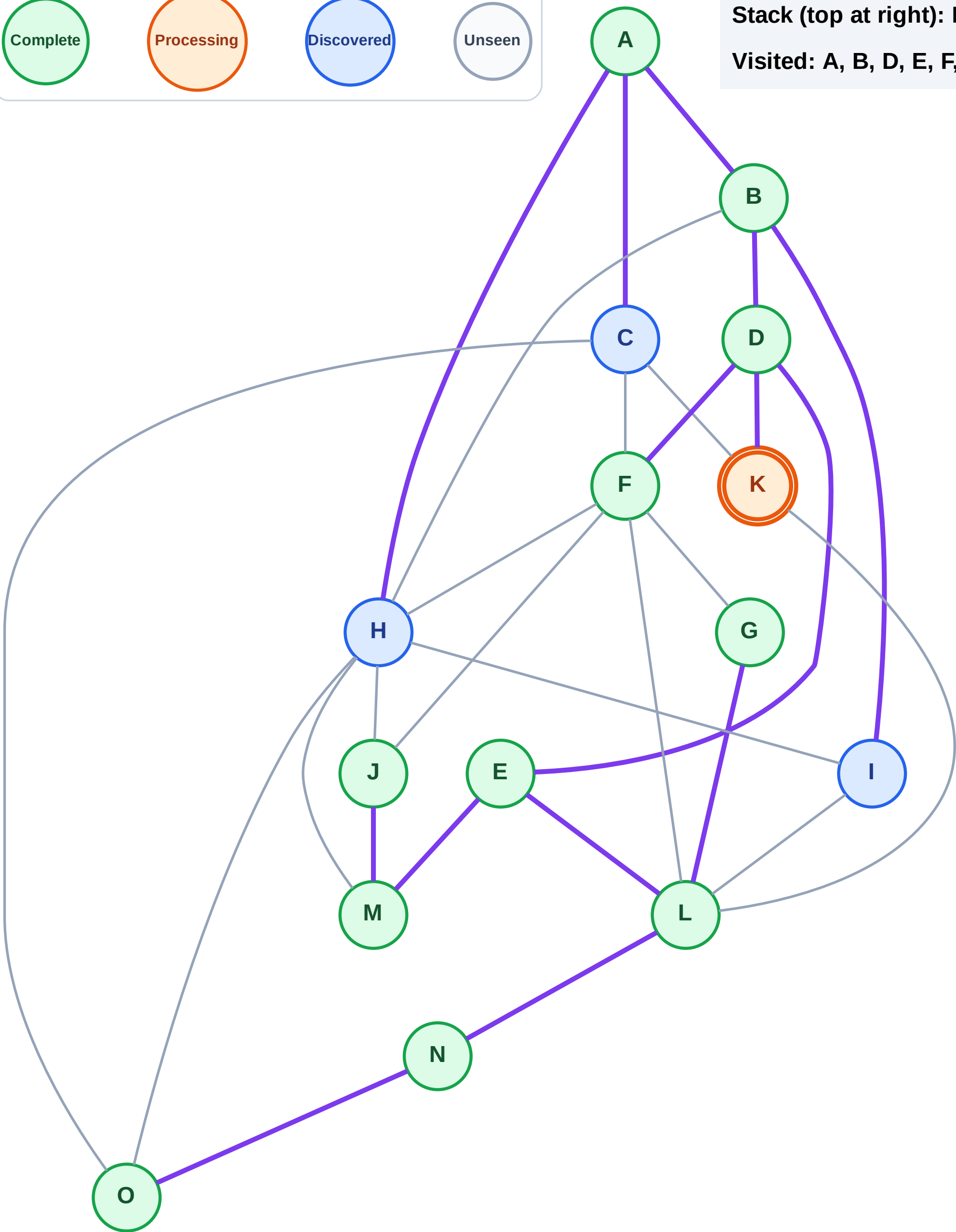
Step 24: Process node K

Legend



Stack (top at right): H | C | I

Visited: A, B, D, E, F, G, J, L, M, N, O



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I

Visited: A, B, D, E, F, G, J, K, L, M, N, O

